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Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels, Part II: Active cooling techniques

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ABSTRACT

This paper deals with active cooling techniques for photovoltaic panels (PVs) where a detailed review was obtained as well as analysis by examining the findings of existing literature. Based on the obtained review, an elaboration of the main performance parameters was obtained for each specific considered coolant (air, water or nanofluids). It was found that the less investigated cooling techniques are the ones related to CPV (Concentrated Photovoltaic) systems and only a few studies exist with nanofluids as the considered coolant. The majority of tested active cooling options are based on water as the coolant and for PV/T (Photovoltaic/Thermal) configurations. The economic and environmental aspects of the active cooling techniques were not analysed in the majority of research studies thus there is an obvious gap in the existing literature. Therefore, the main outcomes of the herein obtained research are reflected through summarized analysis of all important aspects related to active cooling options for PV applications (performance, economic and environmental). According to the obtained results, the highest increase in PV panel performance is achieved by water base cooling techniques and they range from about 10% to 20% on average. To analyse economic and environmental aspects, a 30 kW PV system was studied as a case study. Regarding the economic aspect, the LCOE (Levelized Cost of Electricity) for the considered case study of a 30 kW PV system ranged from 0.096 €/kW h to 0.159 €/kW h. For the given circumstances, it was found that the considered active based cooling options were not economically viable (it is crucial to ensure optimization for the specific liquid (water) based active cooling technique followed by "smart regulation" in order to provide a more reasonable LCOE). However, with the proper optimization of active cooling techniques, it is more than reasonable to expect an additional LCOE reduction as it could significantly reduce the operating cost. The environmental analysis (LCA) showed that out of all the herein evaluated cooling techniques, the air based cooling techniques are the most harmful to the environment which is primarily due to more intense global warming and environmental acidification effect. Other environmental impacts are approximately of the same magnitude for the specific analysed active cooling options.

1. Introduction

The siliceous based photovoltaic (PV) technology is currently the most dominant PV technology on the market, although it is the oldest photovoltaic technology [1]. Other PV technologies such as amorphous, CdTe, CIS/CIGS, did not find a significant market share which is mostly due to their lower average energy conversion efficiencies, usually from 6% to max. 12% (when compared to Si-mono or Si-poly PV technologies), as well as other issues related to their implementation in practical terms. However, there are certain applications where other markets available technologies (as Thin Film) are more suitable than convectional Si-based. In recent years, novel market PV technologies have appeared on the market (i.e. bifacial PV panels [2] and HIT PV panels [3]). The main issues related to the relatively novel PV technologies are either their limitation (applicability) for specific applications or their general high investment. Currently, the highest recorded energy conversion efficiency for market available PV technologies is in the case of HIT PV panels which equals to about 25.6% [3] (when CPV systems are excluded as they reach the highest efficiency levels, i.e. more than 40%, Fig. 1). On the other side, convectional and most used Si-based PV technologies have an average energy conversion efficiency that usually ranges in practical terms from about 10% to 15%, (depending from the specific geographical location). However, an analysis obtained by Fraunhofer ISE [4], emphasizes that there is a certain progress related

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Nomenclature

AC	total life cycle cost, €
CRF	capital recovery factor,
EO	average annual overall energy output from the hybrid energy system, kW h/year
IC	installation cost (overall investment), €
n	amortization period, years
OM	operation and maintenance cost, €/year
p	interest rate,% p.a.

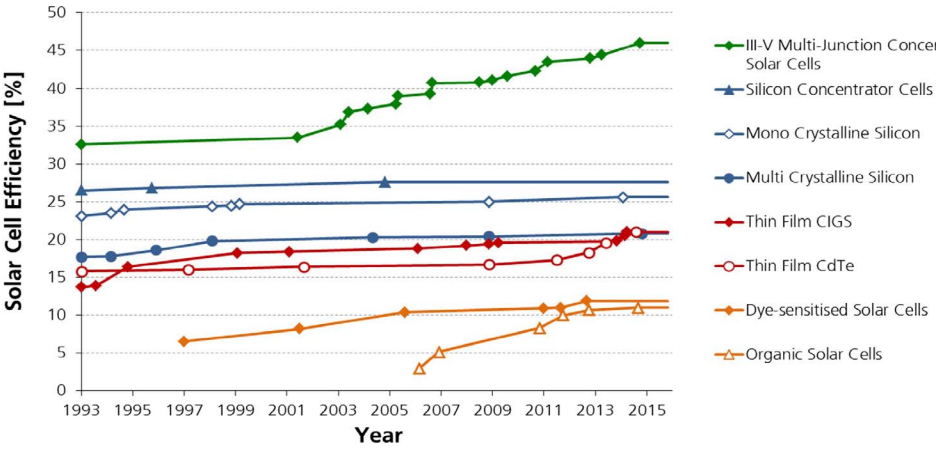


Fig. 1. Laboratory efficiency of different PV technologies [4].

to the increase in average efficiency regarding silicon PV technology, achieved by the further development and optimization of its technology (increase to about 17% and about 21% for super mono). With the development of technology, there is a significant reduction of material usage (more than 2.5 times [4]).

The main disadvantage of the most used silicone based PV technology is its sensitivity to the operating temperature magnitude. Namely, the energy conversion efficiency is reduced with the increase of the PV panel operating temperature which usually ranges from 0.4%/°C to 0.5%/°C (less for amorphous siliceous technology, about 0.25%/°C). Therefore, by limiting the PV panel operating temperature, a higher energy efficiency rate could be ensured. The previous reason resulted in significant research efforts in past years and focused on the development of efficient cooling techniques for PV panels. In general, cooling techniques for PVs can be divided into active and passive ones. The herein provided study is focused on active cooling techniques and different active cooling options were generally investigated regarding existing research studies which are related to different applications (PV, PV/T, BIPV, CPV, etc.). Thus, besides the development of novel photovoltaic technologies, an investigation of thermal phenomena as well as novel potential PV applications is also crucial issues that are under research activity [5–8].

A review of the advances in liquid based photovoltaic PV/T collectors was obtained in [8]. The authors analysed water based and refrigerant based PV/T systems, where the mentioned fluids are generally used to remove sufficient heat from PV/T panels. The main influential parameters were summarized and presented, which were related to general PV/T system performance. It was found that PV/T systems still need significant development in order to obtain reasonable efficiency, and that different cooling strategies need to be proposed and analysed with respect to the specific climate conditions. According to authors, the main direction for the development of PV/T systems should be the integration of a combined PV/T water cooled system for space heating and cooling applications (in order to develop a multi-functional energy system, where COP (Coefficient of Performance) could be significantly improved when compared to conventional heat pump systems).

A review of cooling techniques for a variety of solar collectors and PV cells was obtained in [9] and related to the application of nanofluids. A general overview of the cooling techniques was elaborated from a performance aspect as well as from an environmental aspect (where solar collectors as well as PV and PV/T configurations were analysed). A detailed heat transfer analysis related to nanofluids was also obtained for a variety of experimental configurations (flat plate solar collectors, direct absorption solar collectors, evacuated tube solar collectors, parabolic through, concentrated-parabolic and finally PV/T collectors). An exergy and energy analysis was also performed and annual thermal energy and exergy gains were calculated for different technologies. Finally, the authors concluded that the application of nanofulids could be reasonable in the future, especially for PV/T systems (however, only if a substantial improvement in efficiency and a reduction in costs could be ensured).

An energy and thermo-fluid dynamics evaluation of the photovoltaic systems cooled by water and air was elaborated in [10]. The analysis was obtained for different cooling techniques using TRNSYS software where the final goal was to check the annual performance of the considered cooling techniques. The cooling performance of air and water based cooling systems was analysed in detail, as well as the energy analysis with respect to different construction solutions. Regarding the water based cooling systems for PVs; it was found that the application of plastic materials was most favourable for a variety of reasons when compared to metal sheets. Further, water based cooling systems are generally more complex in terms of overall installation and other issues. Finally, the authors concluded that the yearly improvement of convectional PV systems with air or water based cooling is expected to be about 5%.

A detailed review of cooling techniques related to solar concentrating photovoltaics (CPV) as well as its recent development was obtained in [11]. Various CPV cooling techniques were addressed in detail as for example: heat pipe cooling, water cooling, jet impingement cooling, CPV/T hydraulic system, liquid immersion cooling, microchannel heat sink cooling, PCM (Phase Change Material) based cooling, thermos-photovoltaic cooling and ground coupled cooling systems. The authors summarized the considered CPV techniques with detailed specifications for each technology (type of cooling system, CPV cell material, CPV cell temperature, cooling effect, and concentration ratio, type of concentrator, flow conditions, heat transfer, location and surrounding conditions). The advantages and disadvantages for each considered cooling technique, and their general challenges in construction and application were addressed. The authors concluded that additional experimental work is still needed in order to determine the best method for CPV system cooling. Namely, the choice of cooling method is extremely sensitive regarding the specific location (climate conditions) and assumed application as concluded in [11].

The main objective of the herein elaborated research study was to obtain a comprehensive analysis and evaluation of active cooling techniques for photovoltaics. According to the obtained evaluation of cooling techniques for PVs, a 30 kW PV system case study was considered in order to calculate the general LCOE as well as to analyse the environmental aspect (LCA-Life Cycle Assessment) for certain, i.e. reasonable cooling options. Beside the performance analysis the main idea was also to consider the general economic and environmental aspects of the cooling techniques as there is a general gap in the existing literature (gap related to economic analysis, as a crucial factor for the active cooling technique consideration).

2. Comprehensive analysis and review of active cooling techniques for photovoltaics

In this section, only relevant references from the last two decades were addressed that are related to different active cooling techniques for PVs. The main idea of this section was to prepare a base for later economic and LCA evaluation of specific, i.e. the most reasonable active cooling techniques.

An active cooling technique for PV panels was proposed and experimentally evaluated in [12]. The proposed technique assumed the application of water flow over the front surface of the PV panel with an average applied water flow of about 4.4 l/min per m^2. The water was distributed by twelve nozzles mounted on the top of the PV module, Fig. 2. The average thickness of the water flow layer was around 1.0 mm, with a specific evaporation rate of about 1.0 l/h per m^2 (total evaporation cooling rate equivalent to 680 W h per hour). The experimental setup was equipped with a tracking system that was manually controlled. With the application of the proposed cooling technique, the operating PV panel temperature was significantly reduced from 60 °C down to 38 °C. The achieved increase in electricity delivery registered over the day was in the amount of 10.3%. Furthermore, the overall reported efficiency increase was from around 10.5% (uncooled panel) to around 12.0% in the case of a cooled PV panel. Its economic aspect was also considered and the considered cooling technique was found not to be economically viable due to the low efficiency of the circulation pumps. However, the authors addressed that it could be possible to ensure a net positive increase in electricity yield with the application of highly efficient pumps in an amount of 8% to 9%. The evaporation aspect was not elaborated in the reported research in great detail which is a critical feature for the proposed active cooling technique (the evaporation effect with respect to the general surrounding conditions). Additionally, it would have also been useful if the authors had obtained additional experiments with highly efficient circulation pumps in order to be able to prove and confirm the economic viability of the proposed cooling technique.

Three different PV/T configurations were experimentally analysed and tested in [13], where an air based active cooling configuration was considered through induced airflow. The specific considered experimental configurations are presented in Fig. 3 and the main idea was to use different applied sheets on the backside surface of the PV panel, i.e. PVT/AIR-TMS and PVT/AIR-FIN and to compare them with the referent PV/T panel, Fig. 3. The nominal power output of the tested PV panel was 46 W with a surface of 0.4 m^2. The applied airflow rate ranged from about 30 m^3/h up to 100 m^3/h and the achieved increase in thermal efficiency ranged from about 25% to 30% (the PVT/AIR-FIN system achieved the maximum). In the case of steady state conditions, the induced airflow ensured a reduction in PV panel operating temperature ranging from 45 °C to 65 °C (depending from the specific PV/T configuration), while the PV panel's referent PV/T system operating temperature ranged from 55 °C to 75 °C. The previous result was reported for solar irradiation levels ranging from 700 W/m^2 to 800 W/m^2. The minimal reduction in PV panel operating temperature was about 5 °C, it is thus reasonable to expect an average reduction of 10 °C regarding the PV panel temperature. There was a recorded power loss of less than 1% when driving the fan units and the authors found this to be negligible. An increase in electrical efficiency was only reported for the referent PV/T system (where it ranged from 9% to 12.8%) and an efficiency increase for the other configuration was not reported. It seems that the thermal aspect was more what the authors focused on, where increases in thermal efficiency were the prime target. In any case, increases in electrical efficiency (or eventually decreases) should be analysed for the proposed PV/T configurations as an integral approach (and finally to be able to obtain an economic evaluation for the proposed cooling technique).

The reported study [14] represents the continuation of the previously elaborated research activity in [13]. Further progress in [14] was related to the development of a numerical model (of their proposed PV/T air photovoltaic collector) with validation by using experimental data (good agreement was achieved between the simulated and measured values). The authors once again did not analyse the electrical efficiency of the considered PV/T configurations and the focus of their research was on PV/T system thermal efficiency improvement as already mentioned when addressing the Ref. [13].

A TESPI concept (Thermal Electric Solar Panel Integration) was proposed and experimentally analysed in [15]. The active cooling system consisted of a convectional PV panel that was equipped with an added polycarbonate box on the top of the PV panel, Fig. 4. Below the polycarbonate box, water circulated with a water layer thickness of about 2.5 cm. The specific thickness was selected based on weight issues and reasonable expected increases in efficiency [16]. The TESPI panel was insulated with a 40 mm polyurethane layer in order to be able to achieve reasonable thermal efficiency, i.e. to reduce heat loss. The proposed concept was tested for Mediterranean climate conditions (Italy, Pisa, latitude of solar irradiation levels ranging from 350 W/m^2 to 650 W/m^2) and specific experimental results are presented in Fig. 5. It was found that the application of a water layer over the front PV panel surface caused an increase in electrical efficiency, when compared with the referent one, i.e. standard PV panel of same size. The maximal increase in electrical efficiency was found to be around 2.0% with an average yearly photovoltaic conversion efficiency of around 13%, Fig. 5. The TESPI system was also able to produce hot water with an average temperature of about 40 °C, while the maximal achieved temperature was 63 °C. The global efficiency of the TESPI system was mostly above 15% during the year but it generally ranged from 15% to 20%. A valuable feature that was not analysed was its economic aspect as a PV/T system was in fact considered and which should be obtained in order to be able to check the economic viability of the examined TESPI concept.

A thermosiphon PV/T system was developed and tested for typical Chinese climate conditions (city Hefei, solar irradiation levels from 300 W/m^2 to 800 W/m^2) [17]. The system was assembled with a



Fig. 2. Detail of water distribution over front PV panel surface [12].

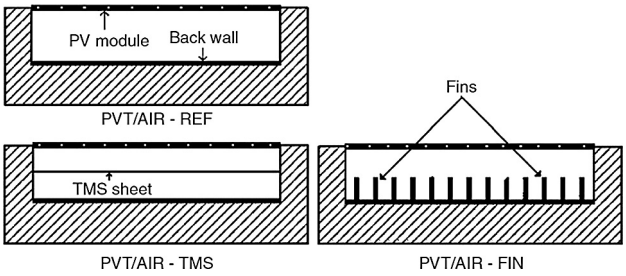


Fig. 3. Characteristic cross-sections of compared PV/T system [13].

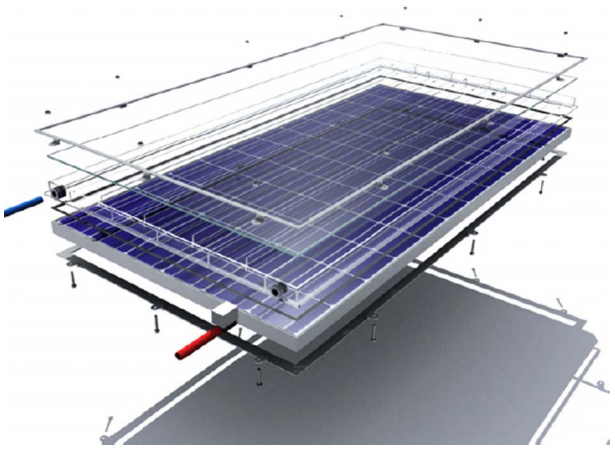


Fig. 4. TESPI concept [15].

monocrystalline PV panel (120 × 120 mm, inclination 35°) and used a modified backside PV panel surface in such a manner that copper tubes were mounted on the aluminium absorber plate and covered with a thermal insulation layer, Fig. 6. The obtained experimental results showed that the proposed cooling configuration had a favourable effect but only regarding the thermal aspect while there was a reduction in electrical efficiency. Namely, the average increase in thermal efficiency was found to be about 40%, while the reduction in electrical efficiency ranged from 10% to 23% on average. The PV string temperature variations were not considered in the obtained experimental approach which could be useful in order to be able to detect the temperature variations over each PV cell. Possible design improvements that would improve electrical efficiency were not discussed, i.e. to avoid degradation when compared to the referent PV panel. The overall investment estimation for the proposed configuration was not addressed as well as the general economic aspect.

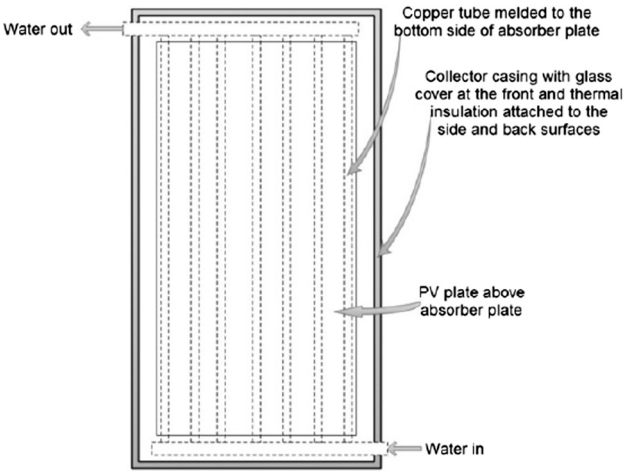


Fig. 6. Experimental configuration of tested PV/T system [17].

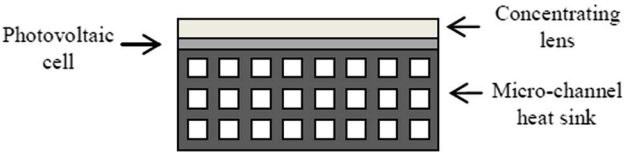


Fig. 7. Simplified sketch of micro channel heat sink [18].

An active cooling system based on a mini and micro channel heat sink was elaborated in [18] and schematically presented in Fig. 7 for the case of a photovoltaic system with concentrating lens. Numerical simulations were obtained for a two-phase flow case and the numerical results were later compared with the available experimental data. Specific pressure drops and heat flux distributions were also examined. The analysis did not include data related to the energy conversion's efficiency improvement and its economic aspect was not addressed (the study was more focused on the flow and thermal phenomena inside the micro channels). The detection of efficiency improvement and experimental validation is crucial for the examined active cooling system.

An air based cooled PV/T configuration was proposed for the cooling of a PV panel through induced airflow on the backside surface of the PV panel, Fig. 8, using a blower fan and specially designed duct system [19]. The system was assembled from four 55 W polycrystalline PV panels and was numerically simulated. It was tested in Singapore climate conditions. The added fins in the duct system enhanced the convective heat transfer and the system was also equipped with a maximum power point tracker to maximize the PV panel's power

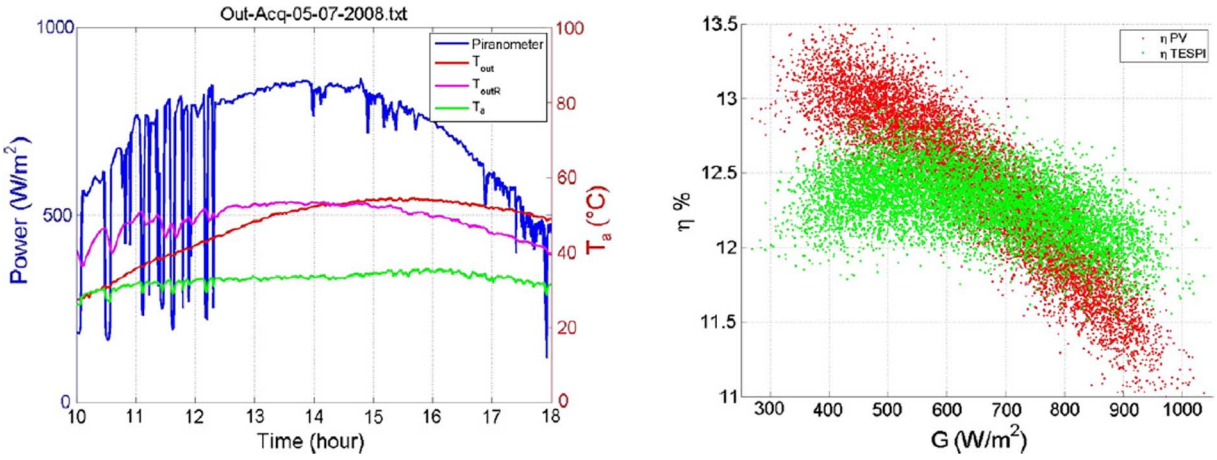


Fig. 5. Characteristic experimental results [15].

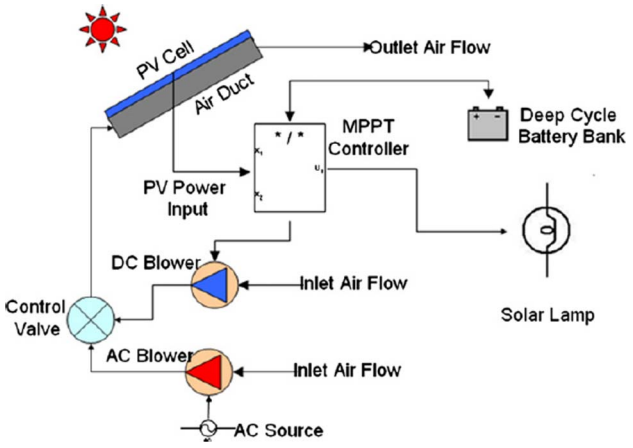


Fig. 8. Simplified sketch of experimental setup [19].

output. The obtained increase in efficiency ranged from 12% to 14% on average with an associated reduction of about 30 °C in PV panel operating temperature and the total examined system efficiency being from 50% to 70%. The efficiency of the non-cooled PV panel ranged from 8% to 9% and a significant improvement in electrical efficiency was achieved. An analysis was performed for optimal airflow in order to detect the maximal heat removed from the PV panel. However, the overall investment cost as well as operating cost for the considered PV/T system was not addressed and it would also be useful to reconsider the additional energy recovery option for the outlet air in the air duct system as additional energy saving could be achieved with a favourable effect on the economic aspect.

A concentrated monocrystalline photovoltaic system (CPV) cooled by water flow was developed and tested in Chinese climate conditions (city of Nanjing) [20]. The specific experimental setup is presented in Fig. 9 and it can be concluded that the system was rather simple as it was assembled from a reflector and water tubes. Solar cells were fixed on the south with an inclination of 35° and the solar intensity during the experiments ranged from 750 W/m^2 to 1019 W/m^2 (with surrounding air temperatures being at around 30 °C and the wind velocity being at around 0.25 m/s). By providing CPV system cooling, it was possible to ensure 4.7–5.2 times higher power output when compared with the fixed system. Furthermore, the average efficiency of the CPV system was about 7.81%, while the efficiency of the fixed PV system was 10.68%. The main reason for the previous was higher operating temperatures in the case of the CPV system (51–61 °C) while in the case of the fixed system, they were from 51 °C to 61 °C. Besides higher operating temperatures, the CPV system was also less efficient due to concentrator geometry imperfections, optical losses and non-uniform solar illumination levels. The water flow effect was also investigated and



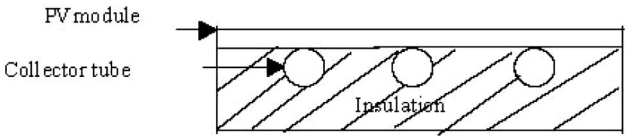


Fig. 10. Experimental setup of PV/T system with schematic [21].

optimal value was found at a water flow rate of about 0.05 kg/s which corresponds with an electrical efficiency of about 8.2%.

The obtained experimental approach schematic was not reported, further, the cooling flow rate was not analysed as an important factor that significantly affects the economic viability of the proposed CPV concept. The optical losses were not analysed and the redesign of the concentrator geometry was not considered, which is crucial in order to be able to achieve improvements in CPV system efficiency.

A novel PV/T system assembled from a polycrystalline PV panel (46 cm × 56.5 cm) with a mounted heat exchanger at the backside surface of the PV panel was proposed in [21] and a specific experimental setup was presented in Fig. 10. The heat exchanger consisted of copper tubes and it was mounted on the copper absorber plate where the whole system was thermally insulated in order to reduce heat losses to the surroundings. The system was experimentally tested in Egyptian climate conditions (Cairo) with solar irradiation levels being from 400 W/m^2 to 850 W/m^2 and fixed at 30°. Finally, the whole system was fitted inside a wooden box, and equipped with an 80 l water storage tank. According to the experimental experience, it was found that it was possible to reduce the PV panel operating temperature to about 10 °C when compared to the non-cooled PV panel and same general circumstances. The average achieved increase in efficiency ranged from 6.8% to 11% for the cooled PV panel case, while for the non-cooled PV panel case, it ranged from 6.2% to 9.0%. The overall reported average efficiency regarding the considered PV/T system was 42.45%, while the average increase in electrical efficiency was about 2%. A clear schematic of the proposed PV/T system was not reported and the effect of water flow magnitude to the overall system efficiency was not analysed.



Fig. 9. Proposed and tested CPV system with detail of the water cooling pipes [20].

Additionally, the operation cost based on the experimental experience, as well as economic aspect was not addressed. A calculation of rejected heat is mandatory to be able to detect the general utilization potential of waste heat.

A novel and rather complex PV/T system was proposed and reported in [22] for the case of building applications, i.e. residential facilities. According to Fig. 11, the system was assembled from an integrated PV array that was cooled by a water circulation system. The water circulation system was designed in two specific configurations, i.e. one was with typical tube connections and the second was a double serpentine connection. The PV/T system prototype was experimentally tested under laboratory conditions and exposed to specific solar irradiation levels at 850 W/m^2 and 1100 W/m^2. The applied water flow ranged from 33 ml/min to 66 ml/min with temperatures being from 32 °C to 34 °C. The decrease in operating PV panel temperatures ranged from 10 °C to 20 °C on average, while the obtained increase in PV panel efficiency ranged from 1.1% to 1.5% (depending from the applied water flow rate and solar irradiation intensity). The highest reached PV panel operating temperature reduction ranged from 50 °C to 55 °C. Based on the experimental experience, it was found that the double serpentine connection enabled a more uniform temperature distribution. However, a single serpentine connection is much easier for tube connection and related to lower initial costs. A detail economic analysis was also reported for the proposed PV/T concept as well as its specified costs. The overall reported cost for the PV/T plant (assembled from 84 modules) was estimated to be around $66,664 (each panel producing about 182 W of electric power output and around 557 W of thermal power output at a solar irradiation level of 850 W/m^2). The overall produced energy LCOE was not calculated and it should be important to check the economic feasibility of the proposed building integrated PV/T concept. Further, the hot water preparation potential was not calculated as well as the effect of water flow magnitude with respect to the overall efficiency of the PV/T configuration.

An active cooling technique for photovoltaic panels that was based on a water circulation system was applied on the backside surface of a PV panel (PVC tubes) [23]. The system consisted of a PV/T panel, water circulation system, water reservoir, Fig. 12 and was tested in hot climate conditions (Dhahran, Saudia Arabia). The average solar irradiation was 710 W/m^2 with a maximal value of 979 W/m^2 and a surrounding air temperature of around 21 °C accompanied with a wind velocity of 1.5 m/s. The system was also equipped with an MPP tracking system to maximize the delivered output from the PV panel. Besides the experimental approach, the system was also tested in a numerical manner using EES software where reasonable matching was found between the measured and simulated values. The water flow rate ranged

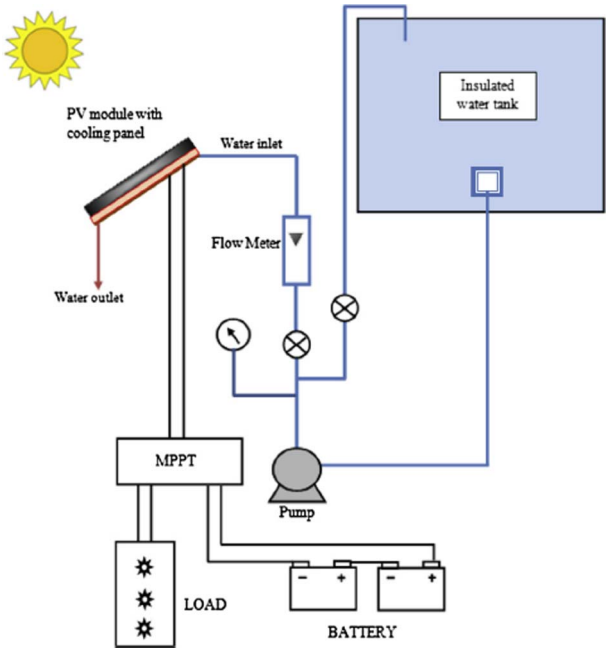


Fig. 12. Schematic design of considered experiment [23].

on average from 0.9 l/min up to 2.52 l/min, while the maximal flow rate was 3.6 l/min. According to the obtained numerical analysis, the average operating PV panel temperature was 30.5 °C for the cooled and 37.8 °C for the non-cooled PV panel. A maximal reduction of the PV panel operating temperature was achieved at the backside surface of the PV panel and it was around 17 °C. Regarding its performance response, the maximal increase in PV panel electrical efficiency ranged from 15.8% to 18.0%. The efficiency of the non-cooled PV panel ranged from 15.2% to 15.7%, while in the case of the cooled PV panel ranged from 16.2% to 18.4%. The maximal power output of the non-cooled PV panel was about 185 W and around 211 W for the cooled, i.e. the average increase in PV power output was 10%. The economic aspect was not addressed for the proposed configuration which is critical in order to check the general suitability of the considered active cooling strategy, i.e. operation and investment cost should be detected. An overall rejected heat calculation from the PV panel to the water was not obtained, i.e. analysis of energy potential for hot water preparation was not investigated and it should be mandatory for PV/T systems.

A water based cooling technique for PV panels was proposed with

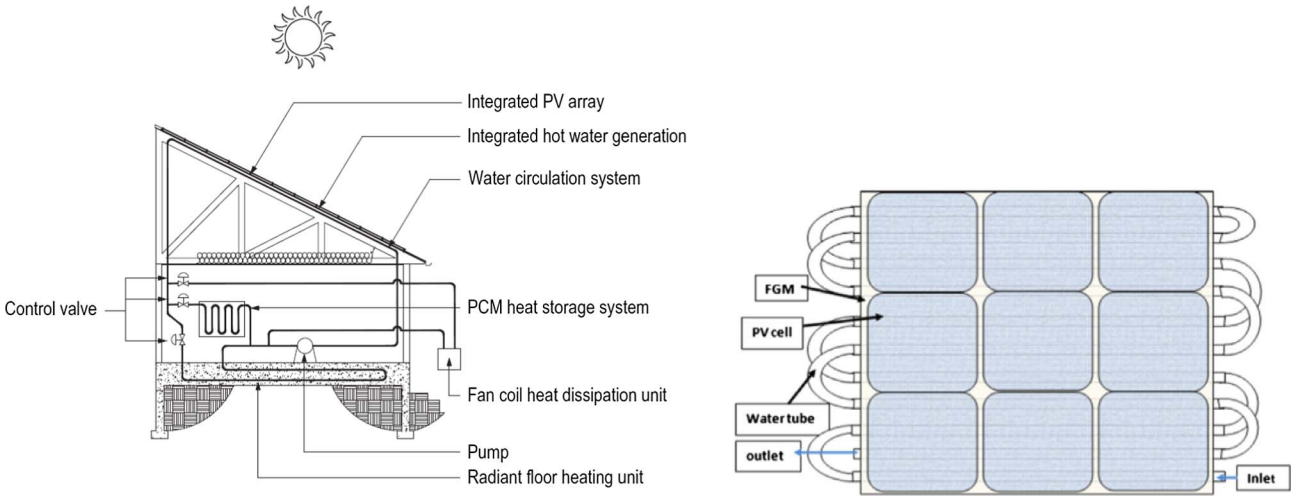


Fig. 11. Proposed PV/T system for residential applications [22].

the application of water flow (water droplets) over the front PV panel surface. The system consisted of six monocrystalline PV panels, each with 185 W in nominal power and a water tank capacity of 0.3 m^3, Fig. 13. The PV system was tested under Cairo climate conditions with the solar irradiation levels ranging from 600 W/m^2 to 800 W/m^2. The main objective of the research was to minimize water waste and electricity for the proposed cooling options [24], where a mathematical model was also developed for the water cooling technique in order to determine the required time to reduce PV panel operating temperatures to the set 35 °C. Water flow (water droplets) was applied over the front surface of the PV panel with a controlled water flow in the amount of 29 l/min (the water cooling temperature was constant and remained at 25 °C). The average cooling rate was found to be around 2.0 °C/min and it was reasonable to activate the cooling system for PV panel operating temperatures at around 45 °C and to reduce it to 35 °C with cooling (this was found as the most favourable scenario in order to have reasonable PV panel electrical efficiency). For the previously proposed configuration, the system was activated each 5 min to enable a decrease in PV panel operating temperatures for the targeted temperature reduction of 10 °C. PV panel electrical efficiency was found to be 12.5% at operating temperatures of 45 °C and 7.5% for 65 °C, while the referent electrical efficiency was 12% at a PV panel operating temperature of 35 °C. The water flow rate variation and PV system performance effect were not considered. The operation cost associated with the water circulation system was not addressed and the economic aspect was not elaborated. The evaporation effect was not examined with respect to the surrounding conditions which can drastically change the evaporation rate.

An experimental investigation related to the performance of a PV/T system assembled from an absorber and tubes mounted on the backside surface of a PV panel, Fig. 14, was reported in [25]. The system consisted of a PV/T panel and referent PV panel with rated maximal power outputs of 190 W that were cooled with water. Experiments were obtained in the Turkish province of Conya with a solar irradiation peak of around 1050 W/m^2 and a tilt angle of 30°. The maximal PV panel operating temperature was 65 °C and it was reduced to 32 °C for an insolation of around 1000 W/m^2. The maximal electrical efficiency of the PV system was 11.5% (the average was 8%) and the PV/T configuration efficiency was 13.6%. The PV/T configuration was found to produce around 10% more electricity when compared to the convectional PV system on a daily basis. The average increase in PV panel electrical efficiency was found to be around 14.6%. The mass flow rate was only examined for tow specific cases, i.e. of 0.030 kg/s and 0.070 kg/s. The PV/T delivered thermal output was not calculated, i.e. to be able to check the waste heat energy potential and besides this, the economic aspect was not addressed for the proposed configuration.

A water based cooling technique was applied on PV panels and consisted of perforated water tubes which ensured a water layer over the front surface of the PV panel [26] in order to enable cooling. The tubes were mounted on the top of the PV panel where the water layer was applied with an average temperature of 24 °C and with an average flow rate of 2 l/min. The system was tested in Romanian climate conditions (Bucharest) where the maximal intensity of solar irradiation ranged from 850 W/m^2 to 1000 W/m^2. The average obtained reduction in PV operating temperature was around 12.5 °C where average temperature difference at the front and backside of the PV panel was 8 °C. It is interesting and important to emphasize that the temperature disturbance of about 2.5 °C was found before the experiments and it was related to the dirt accumulated on the front surface of the PV panel. In general, an increase in electrical efficiency was not reported as well as for the data related to efficiency improvement. The economic aspect and sensitivity analysis related to the magnitude of applied water flow rate with respect to performance response were not addressed.

PV panel cooling was considered with Boehmite nanofulid ($AlOOxH_2O$), i.e. a mix of Boehmite particles with deionized water was used (i.e. the particles were dispersed in the deionized water) to reduce the PV panel operating temperature [27]. The characteristic experimental setup consisted of a small poly-Si PV cell of 5 W, pump circulation system, data logger and solar simulator as the heat source, Fig. 15. The PV cell was exposed to a constant solar irradiation level of 1000 W/m^2 through the application of the solar simulator at a room temperature of 22 °C. Different weight fraction solutions were applied and ranged from 0.01 wt% to 0.5 wt% and the flow rate of the considered coolant was variated with a maximum rate of 400 ml/min. The experimental results showed that the highest reduction in PV operating temperature, relative to the base fluid (i.e. water) was achieved with the highest imposed fluid flow of 400 ml/min and for the lowest weight fraction of 0.01 wt% (i.e. from 59 °C to 34 °C for a laminar flow regime). However, it is also interesting to mention that for the laminar flow rates of less than 150 ml/min, the highest decrease was obtained for the weight fraction of 0.5 wt% (i.e. from 59 °C to 42 °C). The highest average decrease in PV operating temperature was 18 °C (with a 20.57% increase in PV panel power output) for the case of straight channels. A temperature decrease of 24 °C was detected for the helical channels which had a nanofluid concentration of 0.1 wt% (with a 37.67% increase in PV panel power output). Finally, it was also found that the helical channels were more efficient when compared to the straight ones. The specifications related to PV cell efficiency were not reported so it was not possible to detect an increase in PV panel efficiency. The economic aspect was not considered but should be as it is important when considering PV cell cooling with nanofluids.

The application of (SiO_2)/water nanofluid (with weight factors ranging from 1.0% to 3.0%) was examined in [28] in the case of typical PV/T configurations to reduce the PV panel operating temperature and by that obtain an increase in efficiency. The experimental rig is presented in Fig. 16 and consisted of a PV/T panel, storage tank, heat exchanger, circulation pump and pipeline system as well as from components related to PV systems. A referent PV panel was also ensured in order to be able to compare results. The system was tested under climate conditions of Iran (city of Mashhad), with solar irradiation levels ranging from 600 W/m^2 to 1100 W/m^2 (with the surrounding air temperature ranging from 30 °C to 35 °C).

The operating temperature of the referent PV panel ranged from 48 °C to 69 °C, while the PV/T configuration operating cell temperature ranged depending from the applied weight factor. Namely, PV panel operating temperature ranged from 38 °C to 51 °C (for pure water), further from 37 °C to 49 °C (for 1.0 wt%) and finally from 38 °C to 49 °C (for 3.0 wt%). Regarding performance response, the electrical efficiency in pure water ranged from about 6% to 9.2% (pure water), further from 7.0% to 10.1% (for 1.0 wt%), and finally from 7.9% to 11% (for 3.0 wt %). Therefore, the highest achieved increase in PV panel operating temperature was for a weight factor of 3 wt% nanofluid where it is important to emphasize that the applied mass flow rate for the water was 40 L/h and was 30 L/h for the nanofluid (the optimal values that lead to the highest increase in electrical efficiency). The maximal found

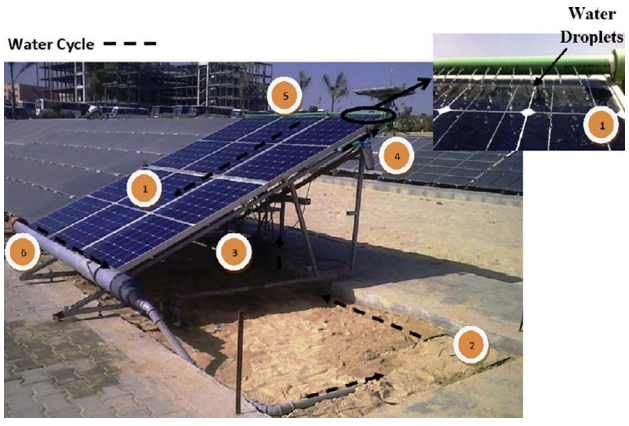


Fig. 13. Experimental setup for proposed water based cooling technique [24].

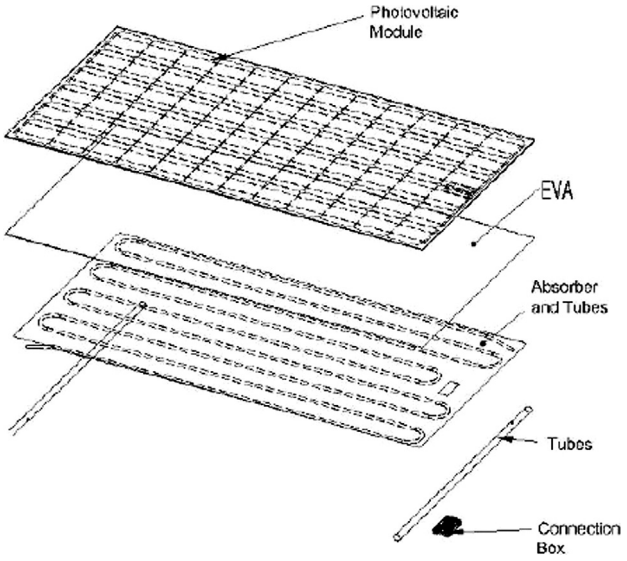


Fig. 14. PV/T system experimentally tested [25].

increase in electrical efficiency was from 9.2% to 11% for 3.0 wt% of nanofluid (with the highest decrease in operating temperature being 19 °C). As in the majority of cases, the economic aspect was not addressed although a nanofluid cooling option was considered (analysis of investment and operation costs for different scenarios related to different applied weight factors.

A self-adjusted jet impingement system was proposed in [29] as an active cooling technique for photovoltaics where a cooling effect was achieved through the application of induced airflow. The schematic of the proposed experimental approach was presented in Fig. 17 where it can be seen that the system was assembled with a specially designed conic tube system using self-adjusted wind vanes and 5 W Poly-Si cells that were tested (with a rated maximal efficiency of 16%). The system was also equipped with an additional wind turbine to act as an additional electricity source (dynamo). A solar simulator was used and the cells were exposed under solar irradiation levels of 570 W/m^2, 910 W/m^2 and 1100 W/m^2. According to the experimental findings, the PV cell operating temperature was stabilized at 57.6 °C, 85.4 °C and 105.4 °C, depending from the applied level of solar irradiation for the case of the non-cooled PV cell. However, with the application of the proposed cooling technique, the average operating temperature of the PV cell was reduced from about 57 °C to 40.4 °C for a solar irradiation level of 510 W/m^2 and air flow rate of 340 m^3/h. It was found that the average PV panel operating temperature was reduced from about 57 °C to 40.4 °C on average due to the cooling effect (reduction rate depends from quantity of air flow). For example, the average reduction of the PV cell operating temperature ranged from 16 °C to 25 °C for an air flow rate of 100 m^3/h and for the above specified solar irradiation levels. The effect of the different air flow magnitudes was also investigated with a constant solar irradiation level of 910 W/m^2, and for an air flow rate of 100 m^3/h to 300 m^3/h, it was found that the maximal achieved increase in PV cell power output was about 12.5%. It was also found that with the addition of a turbine in the system as already mentioned, the overall electricity production was increased by 21% when compared to the cooling system without a turbine. In any case, the proposed approach is more than interesting due to the consideration of an energy recovery system with the application of an additional turbine. However, a crucial feature would be to check the proposed solution on a standard PV panel (higher rated nominal power output) and simultaneously estimate the overall investment and operational cost. Namely, the cooling effect on small PV cells is rather different when compared to mid or large-sized PV panels. Finally, the system should be tested under real circumstances to check the difference between the laboratory results and to get realistic performance output.

A direct liquid-immersion method for CPV systems was examined in [30] using dimethyl silicon oil (viscosity of 2 mm^2/s) to cool the Mon-Si PV cells, each with a single cell power output of 0.14 W and maximal electrical efficiency of 15%.The specific experimental setup is presented in Fig. 18 and the system consisted of a PV panel with a concentrator and appropriate cooling rig using a heat exchanger (overall heat exchange surface area of around 0.8 m^2 between silicon and cooling

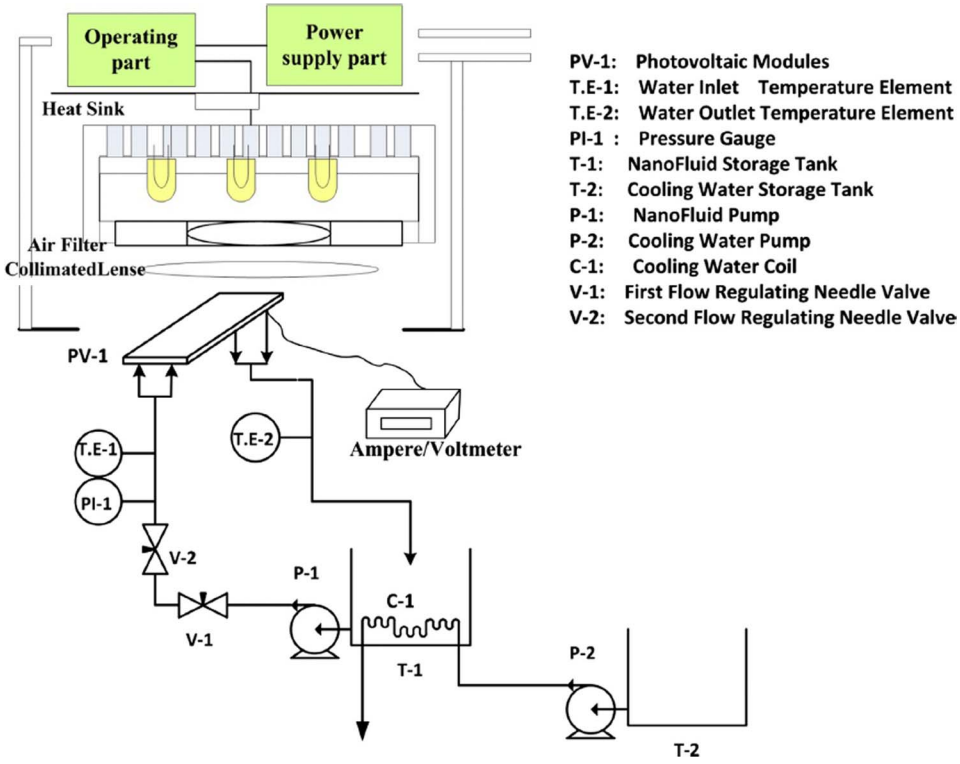


Fig. 15. Experimental setup [27].

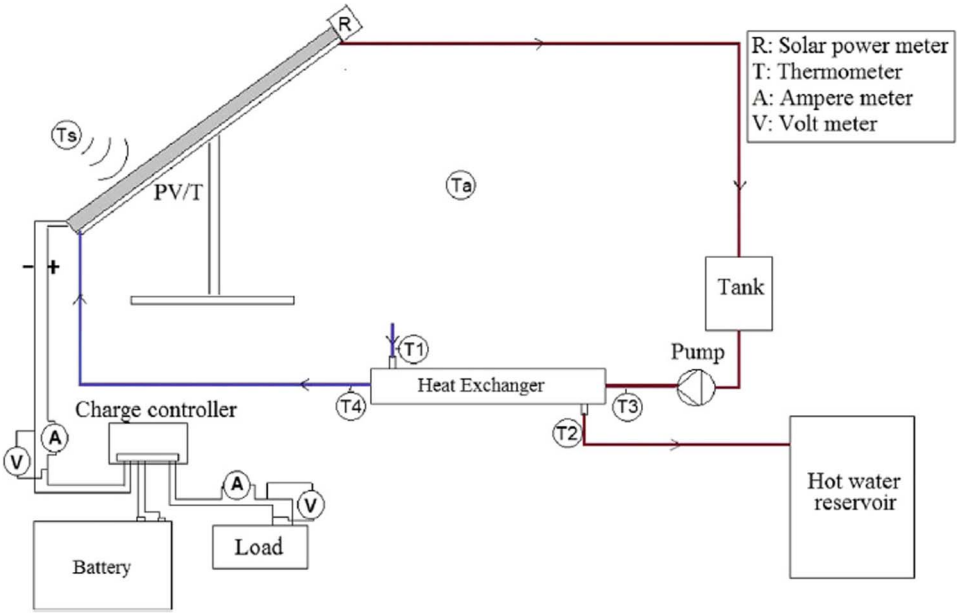


Fig. 16. Schematic design of experimental setup with nanofuid SiO_2/water [28].

water) and circulation pump system. The system was tested for solar irradiation levels that ranged from 700 W/^{2} to 925 W/m^2 with associated coolant flow rates that ranged from 0.22 to 1.09 L/s. It was found that the average cell temperature was linearly dependent from the silicon oil inlet temperature. Namely, for a solar irradiation level of 920 W/m^2, the average cell temperature was maintained from 20 °C to 31 °C and for an inlet oil temperature of 15 °C. Electrical efficiency before the silicon immersion was 13.94% with an inlet silicon temperature of 10.6 °C. The average electrical efficiency was around 13.5% with an inlet silicon temperature of 15 °C, and a degradation of electrical efficiency was present after 270 days, but not significant according to the authors (degradation ranged from 13.94% to 12.76%). The flow rate effect on performance response was not considered but should be important to detect. Further, the operation and overall investment costs were not analysed. It would also be useful to analyse the waste heat potential for potential hot water preparation regarding the proposed system which could be a favourable affect with respect to the economic aspect.

A novel PV/T heat pipe system was proposed in [31], Fig. 19 and assembled from two cooling jackets that were mounted on a 20 W monocrystalline PV panel (with a nominal efficiency of 13.88% and tilt of 40°). The system was cooled by water and the novelty was in the proposed double cooling jackets which boost the cooling effect and is equipped with an MPPT. The PV/T heat pipe system was tested under climate conditions of Iran (Mashad) with solar irradiation levels being around 990 W/m^2 up to 1080 W/m^2 with surrounding air temperatures of 26.5 °C in spring and 36.9 °C in summer. According to the provided experimental results, the average achieved decrease in PV operating temperature was 13 °C and the average increase in electric efficiency was around 9%. Furthermore, the considered system was able to deliver more power output when compared to a convectional PV panel in the amount of 0.72% in spring and 0.88% in summer period. The total exergy efficiency of the proposed system was also obtained. The water flow rate effect and potential for hot water preparation were not examined which is mandatory when considering PV/T systems. The economic evaluation is also missing and was not addressed. In any case, the proposed cooling configuration should be tested on large PV panels in order to be able to check the circumstances for overall performance response and economic viability.

An active cooling system was experimentally analysed in the case of a large scale PV system consisted of 28 monocrystalline PV panels each being of 175 W in nominal power output and tilted at an angle of 30°

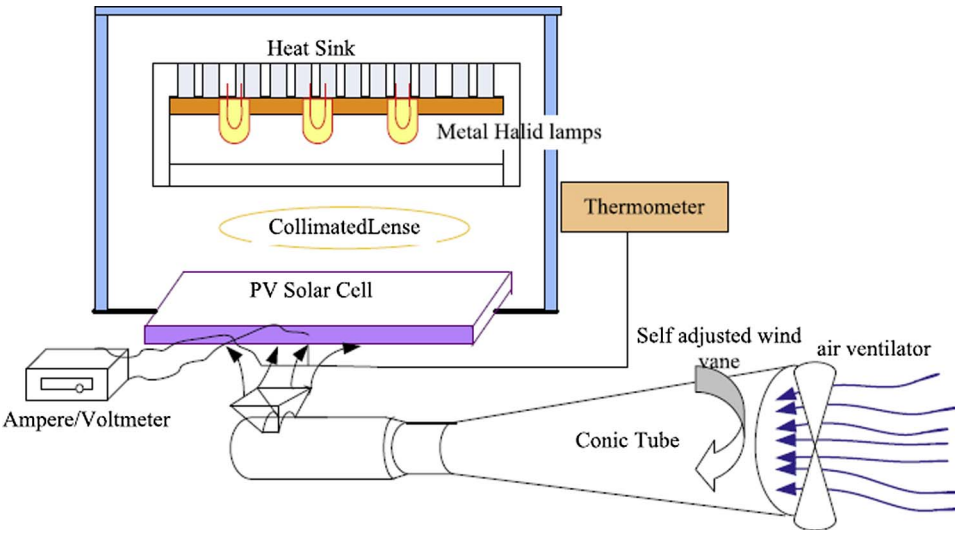


Fig. 17. Schematic design of self-adjusted jet impingement system for PV cooling [29]

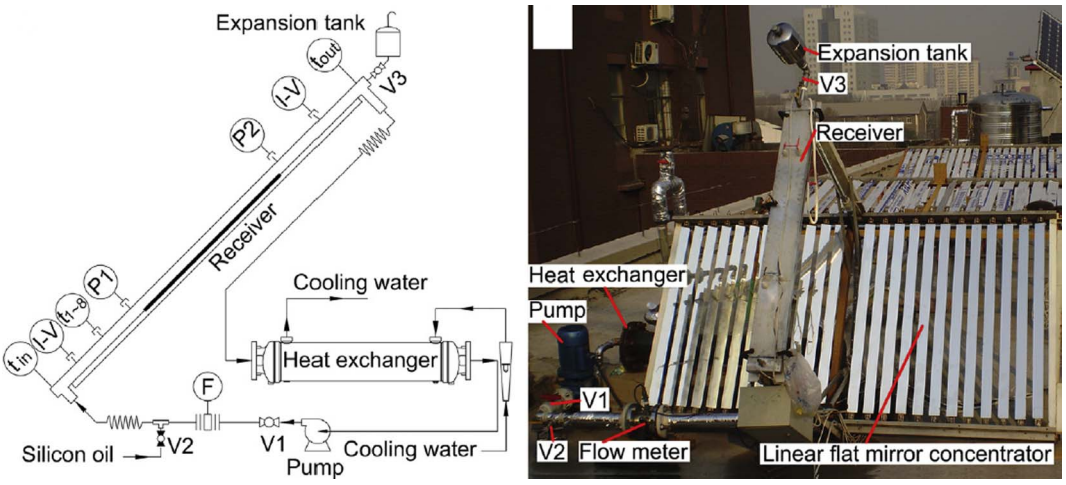


Fig. 18. Liquid-immersion CPV system [30].

[32]. The cooling technique consisted of PVC pipes that were mounted on the top of the PV panels and perforated to disperse water over the panels with an approximate water flow rate of 7 l/min (the pump consumed around 7 W of power output for the circulation system). The system was tested under USA climate conditions (Portland region). Two different PV system rigs were tested, i.e. schematically presented in Fig. 20. The novelty in their approach was that they considered different cooling circumstances as well as different PV configurations. Namely, chilled water was considered with an applied water temperature low of 2.5 °C. Further, the setup with concentrating mirrors (V-trough) was also used to combine the cooling and concentrating technique (the mirrors were obtained from highly reflective Mylar film applied on each side of the PV panel). According to the obtained experiments, the average PV system electricity production was 12.1%, however, the net increase was about 6% (with included pump work for the circulation system). The peak increase for the PV panel power output was 12.9% and it averaged about 10%, i.e. net 5%. Certain PV panels were insulated in order to simulate circumstances where air-cooling was disabled and the delivered PV panel output improvement was 14%, i.e. net 8.3%. Further, when ice water was used for cooling the PVs, the lowest achieved PV panel operating temperature was about 16 °C, with surrounding air temperatures ranging from 17 °C to 19 °C and with a cooling water temperature of 2.5 °C (the average temperature of the non-cooled PV panel was 55 °C). In the previous case, the improvement in PV power output was the largest and in the amount of 24%. Finally, for the case with the V-trough, the peak increase in PV

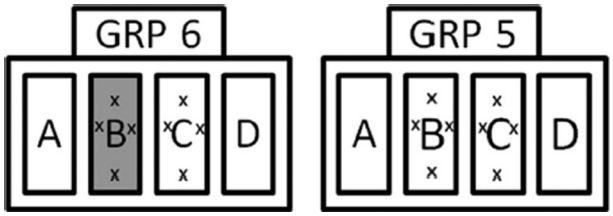


Fig. 20. Simplified schematic design of experimental setup [32].

panel power output was 43%. An overall cost was estimated for the proposed water based cooling technique and the overall investment for the cooling system was estimated to be around $15 per panel. Therefore, for a typical 3 kW home installation, an additional 150 kW h/per year of electricity could be obtained (which would be sufficient to return the invested funds). However, a detailed overall cost specification for the used equipment in the proposed experimental approach was not reported. The specific water cooling temperature for each configuration was not reported and correlated directly with the efficiency improvement. The waste heat potential analysis was not considered as well as the evaporation effect for different surrounding circumstances.

A novel ceiling ventilation system where PV/T panels were integrated with PCM materials was reported in [33]. A simplified concept of an enhanced ceiling ventilation system was presented in Fig. 21, which in fact consisted of a PV panel that was cooled on the backside by airflow through an air channel (the whole system was thermally



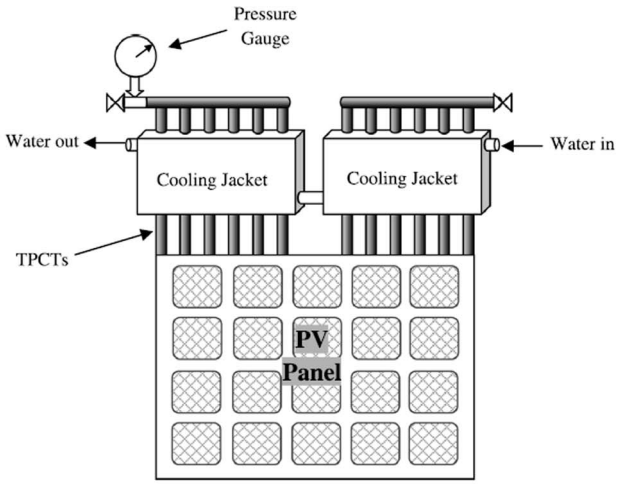


Fig. 19. Heat pipe PV/T system (without thermal insulation) [31].

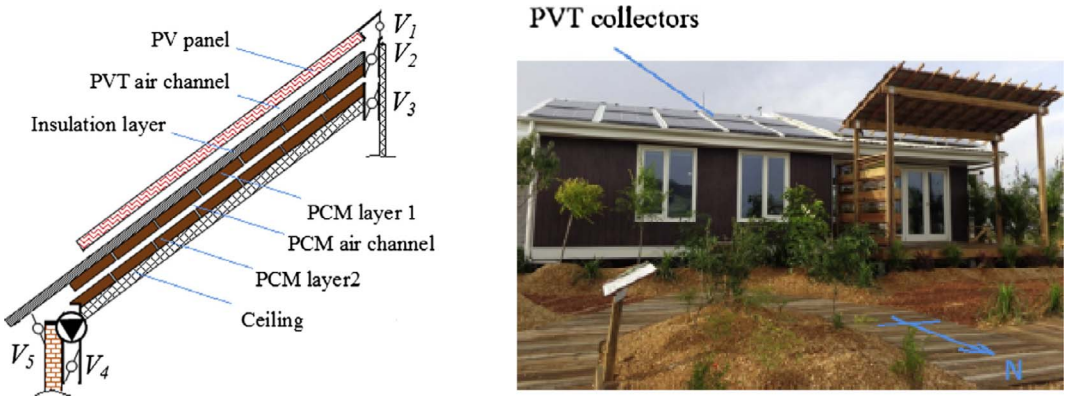


Fig. 21. Simplified sketch of novel ceiling ventilation system with PVT collectors and building facility [33].

insulated) and finally two PCM layers were added before the ceiling (with an ensured air channel between the two PCM layers). A prototype was installed in a building facility with an overall surface of 42 m^2 where the total PV/T collector area was 40 m^2. The used PCM material was SP24E with a thickness of 32 mm and with a melting temperature ranging from 21 °C to 25 °C. Besides the experimental approach, the whole concept was also simulated using TRNSYS software. The system was tested under Australian climate conditions (Sydney). Based on the experimental experience, it was found that the peak PV power output was 1.35 kW in winter period and 1.98 kW in summer period. The average operating PV panel temperature was 24.6 °C in winter period (maximal 44.2 °C) and 42.2 °C in summer period (maximal 71.7 °C). The average electrical efficiency was around 8.3%, both for summer and winter periods. Therefore, it can be concluded that the average reduction in PV panel operating temperature would range between 10 °C and 20 °C, depending from the period of year (with a mean airflow rate of about 500 kg/h for the ceiling ventilation system). A comparison of the proposed PV/T configuration with the referent PV panel was not obtained which should be mandatory in order to check improvement, or even degradation when considering PV system efficiency. Additionally, the economic aspect is also missing, i.e. the cost-benefit analysis of the examined and proposed configuration.

An active cooling technique for photovoltaics suitable for application in residential houses and with considered rainwater utilization was reported in [34]. The system is schematically presented in Fig. 22 and consisted of a PV system (cooled with water applied over the front PV panel surface), water storage tank and sufficient expansion tank to enable water circulation. The novelty of the system is in the application of rainwater and enabled circulation through the expansion tank. The examined PV system had a rated maximal power output of 250 W with a maximal efficiency of 15.4% and PV panel area of 1.46 m^2. The system was simulated for UK climate conditions where solar irradiation ranged from 380 W/m^2 to 798 W/m^2 with air temperatures being from 16 to 27 °C and with associated wind velocities from 4.3 m/s to 10.3 m/s. The system was not tested under realistic circumstances and was just elaborated from a theoretical point of view. It was found that about 175 l/day of rainwater would be available for specific UK conditions. The performance analysis showed that the maximal achieved reduction in PV panel operating temperature was 19 °C and for other circumstances it ranged from 12.5 °C to 18.5 °C. The electrical efficiency of the cooled PV panel was maintained over 14.5% on average, while for the non-cooled PV panel it was below 13.5%. The average obtained increase in PV panel electrical efficiency was about 8.3% and the maximal increase in PV panel power output was around 15%. It was found that the average monthly required quantity of rainwater was less when compared to the collected quantity of rainwater. Basic economic analysis was also obtained for the considered cooling system and it was found that the system would produce around 33.4 kW h per year of additional electricity and that the overall payback period would be around 14 years. The main issue with the previously addressed approach is that the authors did not obtain the experimental validation of their calculations (simulations) as the economic aspect could have been critical for the assumed configuration.

An experimental and numerical analysis of a cooling technique based on the application of a converging channel heat exchanger was elaborated in [35]. The system consisted of eight solar cells connected in series (each of 3.42 W nominal power), an installed MPPT device to maximize the PV power output and with measuring equipment, Fig. 23. The heat exchanger was applied on the backside surface of the PV panel and insulated with glass wool in order to reduce heat transfer losses. System modelling, i.e. CDF analysis was obtained using the FLUENT 6.3 software and the results were in good agreement with the experimental

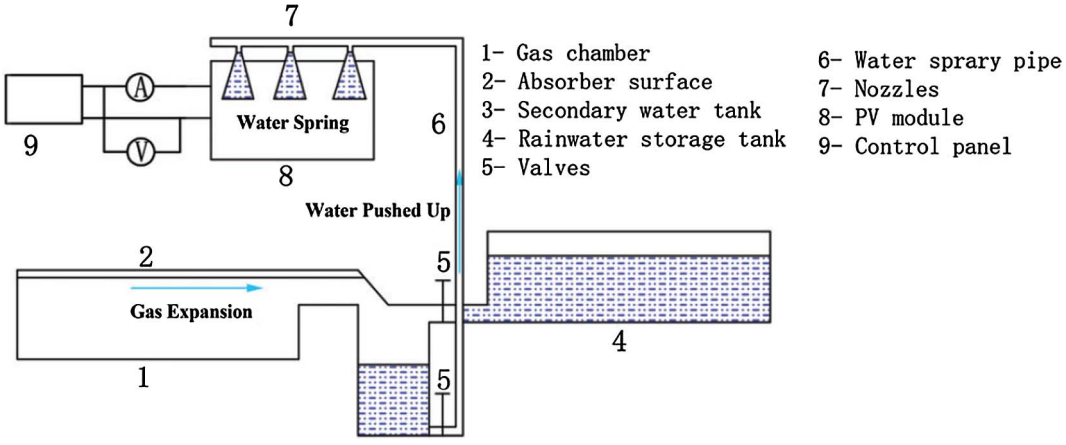


Fig. 22. Schematic design of proposed active cooling for domestic applications [34].

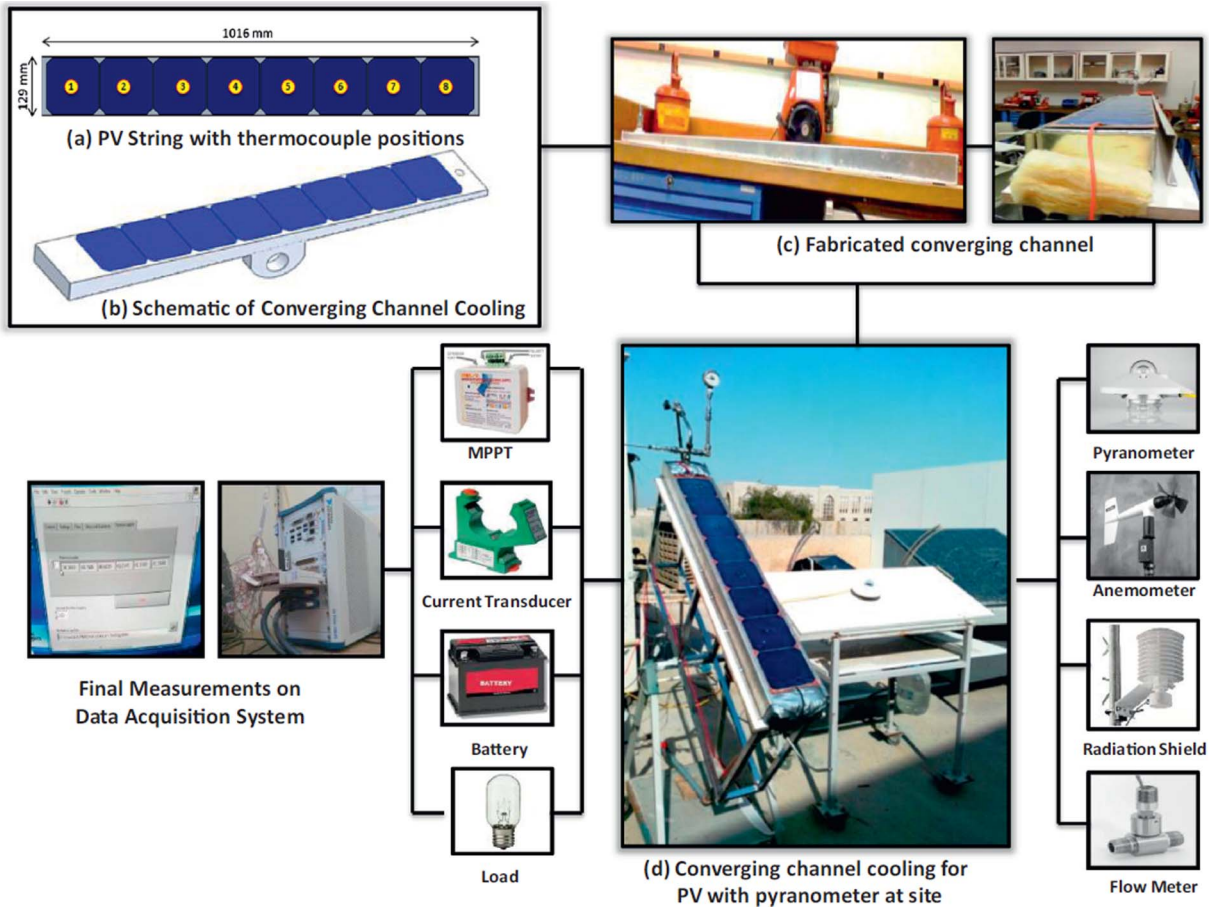


Fig. 23. Experimental setup of cooling technique with converging channel [35].

findings. The system was tested under Saudi Arabian climate conditions (city of Dhahran) with solar irradiation levels ranging from 400 W/m^2 to 960 W/m^2. The system was tested both in winter and summer operations with the summer air temperatures being from 40 °C to 45 °C, and winter from 20 °C to 25 °C. Based on the experimental work, it was found that the PV panel operating temperature was reduced from 71.2 °C to 45.1 °C in summer operation and from 48.3 °C to 36.4 °C in winter operation, i.e. when compared to a non-cooled PV panel. The increase in PV panel power output and electrical efficiency was 35.5% and 27.5% for summer operation, i.e. 26.1% and 36.1% for winter operation. A detailed economic analysis of the proposed cooling technique was also obtained and the LCOE was about 1.59 €/kW h for the non-cooled PV panel to 1.57 €/kW h for the proposed cooling technique case. The system was only tested for one converging channel and it would be interesting to check the same approach with the connection of a few converging channels, i.e. to check the performance response for these circumstances.

A double active cooling technique was examined in [36] which consisted of applied water spray over the front surface of the PV panel (20 W, polycrystalline) and direct water cooling (water flow) on the backside surface of the PV panel, Fig. 24. The system was tested under UAE (Sharjah) climate conditions for solar irradiation levels that ranged from 849 W/m^2 to 863 W/m^2. The operating temperature of the non-cooled PV panel was 42.1 °C and the operating PV panel temperature was reduced from 0.7 °C to 7.7 °C because of the specific cooling technique application. The less efficient cooling scenario was the one related to the cooling of the backside only and the most efficient one related to double cooling. The efficiency improvement ranged from 2.1% to about 4.0% for the case of double cooling (front and backside surface of the PV panel). However, the detailed schematics of the examined active cooling configuration were not reported as well as for the evaporation effect. Furthermore, the economic aspect was also not addressed, i.e. estimation of overall investment and operation cost in particular.

A water spray cooling technique was applied over a 50 W monocrystalline PV panel both on the front and backside of the PV panel



Fig. 24. Experimental setup for proposed double active cooling technique [36].

[37]. The experimental configuration consisted of water nozzles mounted on the front and backside surface of the PV panel that were connected to the water distribution system, Fig. 25. The system was tested under typical Mediterranean climate conditions during periods of highest solar irradiation levels, where the insolation ranged from 810 W/m^2 to 850 W/m^2 (with surrounding air temperatures ranging from 27 °C to 30 °C and the air velocity being under 1.0 m/s). The water spray technique was tested in a few cooling regimes, i.e. front cooling only, backside cooling only and simultaneous front and backside cooling (the water spray was about 17 °C). The best result was obtained for simultaneous cooling where the highest PV panel operating temperature reduction was obtained in the amount of 30 °C (from 56 °C to about 24 °C). The overall PV panel electrical efficiency improvement was 14.3%, however, net improvement with included power loss for the water circulation system was about 5.9%. The PV panel operating temperature reduction ranged from 23 °C to 32 °C with associated efficiency improvement from 12% to 14.3% (net improvement from 3.6% to 5.9%). PV panel power output improvement ranged from 14.0% to 16.0% and net improvement from 5.4% to 7.7%. The proposed water spray technique was also evaluated from an economic point of view and it showed to be economically viable, i.e. more electricity was produced than consumed work for the circulation system. Additionally, a feature that could have been done was the analysis of the evaporation effect as a water spray cooling technique and its overall impact on the economic aspect. A variable water spray temperature effect was also not analysed (a constant water spray temperature was applied in the experimental approach) which could be useful in order to check the performance response with respect to economic viability.

A water and refrigerant based cooling for PVs was analysed from both technical and economic aspects for the climate conditions in [38] where water taken from a well was used as a coolant and which was a novelty in the proposed approach. A schematic overview of the experimental setup was presented in Fig. 26 where it can be noticed that the PV system (Mono-Si with nominal power of 50 W) was cooled by water from a well (temperature average of 17 °C to 19 °C) and the system was equipped with measuring elements as well as with a water distribution system. The system was tested under Hungarian climate conditions and the PV panel temperature was constantly maintained at 25 °C (PV panel cooling was obtained through water sprinkling). Data and statistical analysis was obtained by using the ANOVA and SPSS 22 framework. According to the obtained experimental results, the electrical efficiency improvement was found to be about 12% on average with an associated water consumption of around 22.5 L/. The refrigerant based method showed not to be economically viable and the authors concluded that it was not appropriate for climate conditions of Hungary. The maximal reduction in PV panel operating temperature was about 26 °C and the increase in PV panel output was about 42.8% (average 19% to 25%). The average electric energy produced for the controlled panel was 196 W h and 229 W h for the one cooled with sprinkling water from 9 am to 17 pm (surplus of 22.9 W h, while for the refrigerant based cooling a deficit of −18 W was obtained). The authors stated that the issue that can be found with sprinkling technology is the possible precipitation of inorganic materials which can cause a reduction regarding incoming irradiation. Therefore, the long term effect of possible inorganic material accumulation should be examined and an overall economic evaluation should be obtained (as only a partial economic aspect was addressed).

A novel active cooling technique, i.e. a micro heat pipe array was considered in [39] where two cooling options were analysed for the 10 W PV panel. The first cooling option was active water cooling and the second was passive cooling with air-cooling and the system is schematically presented in Fig. 27. It can be seen in Fig. 27 that the first configuration had ensured air-cooling with an integrated heat pipe, while the second configuration was a water cooled PV panel (backside surface) with an integrated water storage tank. The system was tested under laboratory conditions for an applied solar irradiation level of 1000 W/m^2, with a surrounding air temperature of 25 °C and a tilt angle of 45°. The maximal reported difference of 2.6% in photoelectric conversion was for air and 3.0% for water cooling. The maximal reduction in PV panel operating temperature was 4.7 °C for the case of air cooling and 8 °C for the case of water cooling. The heat pipe cooling option which was cooled by water ensured an average increase of around 9% in the delivered PV panel output. The water flow effect as well as air flow rate was not investigated with respect to PV panel performance. The economic evaluation was also not considered for both examined and proposed configurations.

A different cooling options was experimentally examined for a 40 W Si-poly PV panel as presented in Fig. 28. Three different PV panels were used in the experiment, i.e. the first one being referent, the second with a PCM material (RT42 with about 10.2 kg) and a PV-PCM water system as the third, i.e. PCM layer cooled by water flow (water enabled from water tank). The system was tested under UAE climate conditions (city of Al Ain) with the average solar irradiation values being between 550 W/m^2 and 590 W/m^2 as well as with the peak values being over 900 W/m^2. The experimental results showed that the average temperature drop was about 10 °C for the PV-PCM-water system with an associated 6% average increase in PV power output (and an increase in PV electrical efficiency of about 1%). The economic viability of the active cooling option was also examined and the overall increase in produced electricity was about 18 W h/day (with a required total

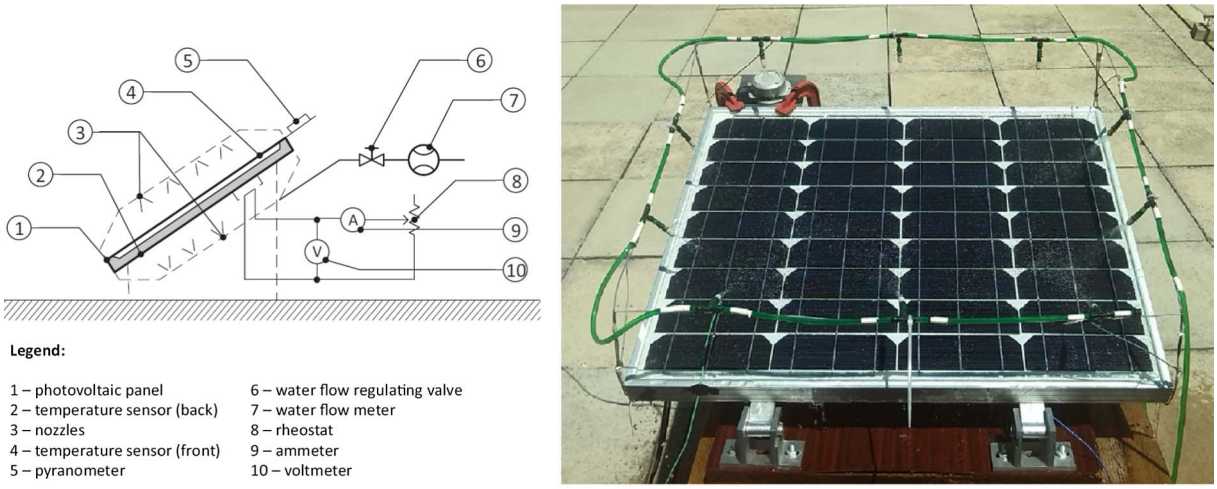


Fig. 25. Water spray cooling technique [37].



Fig. 26. Characteristic experimental setup for water-cooling of PV panel [38].

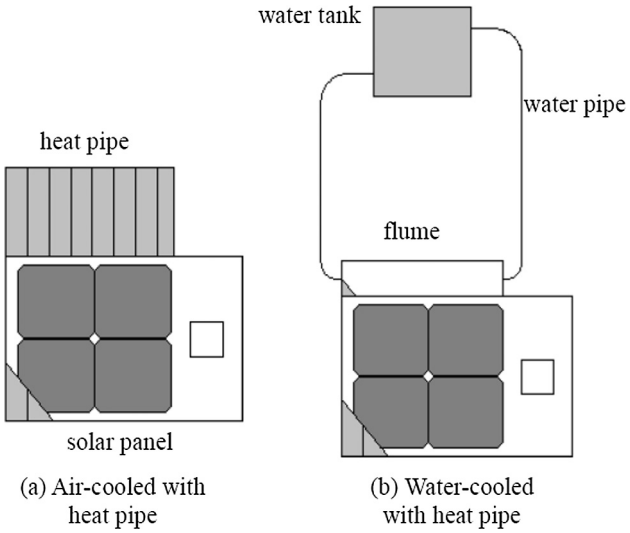


Fig. 27. Proposed cooling concept-micro heat pipe array [39].

consumed energy of 12 W h/day for the water circulation system). However, the overall cost was not addressed, i.e. the analysis of LCOE based on the possible utilization of waste heat along with the water flow rate effect which was also not obtained (see Fig. 29).

An active cooling technique for photovoltaic panels using forced air flow induced by a fan at the backside surface of a PV panel was proposed in [41]. The system was assembled from a blower attached on the specially designed duct system where the air was directed to the backside surface of the PV panel in order to reduce the PV panel operating temperature, Fig. 29. Different polycrystalline PV panels were tested with nominal power outputs ranging from 240 to 280 W. The system was tested in Spanish climate conditions (Cartagena) where the solar irradiation levels ranged from about 300 W/m^2 to over 900 W/m^2. The applied air velocities ranged from 2 m/s to 4 m/s, and the distance between the PV panel and roof was also variated and examined with respect to PV panel performance (the distance ranged from 0.105 m to 0.165 m). For the case of the forced convection, the highest increase in PV panel temperature was achieved when the highest air velocity (4.0 m/s) was applied along with the highest distance between the PV panel and roof. In the previous case, the reduction in PV panel operating temperature ranged from 10 °C to 16 °C. The maximal improvement in efficiency for the active cooling configuration was around 6.3% for maximal applied wind velocity. The distance importance between the PV panel and roof was addressed and it was generally found that larger aspect ratios are more desirable in order to increase heat rejection from the PV panel for both passive and active examined cooling techniques. However, the electricity consumption of the fans, i.e. general economic viability of the proposed cooling technique was not analysed (i.e. cost benefit analysis which is crucial in order to discover the operational cost and to be able to estimate the maintenance cost as the blower fan could be exposed to different possible surrounding circumstances).

3. Elaboration of considered active cooling techniques and general comparison

According to the obtained review and provided analysis of the active cooling techniques for PVs, a summary is given in Table 1 related to the performance parameters as well as other influential parameters. The values in Table 1 are specified either in the form of a single number (peak value in that case) or as value range, and finally as average value

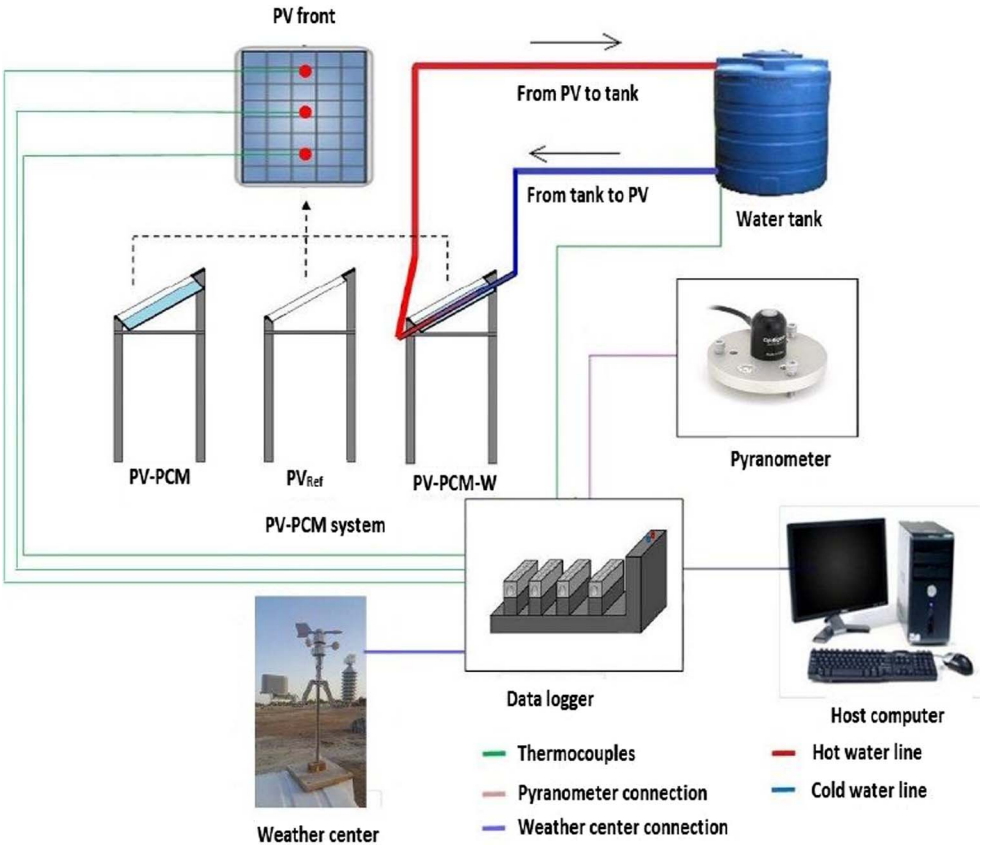


Fig. 28. Active and passive PCM based cooling technique for PVs [40].



Fig. 29. Experimental setup for air based cooling technique, forced convection circulation [41].

(which is marked in the same table). The numbers in Table 1 were gained based on the general availability of the data that were reported in the herein considered references. Further, some data related to the increase in efficiency were not available, so they were calculated based on the available temperature reduction in PV panel operating temperatures for a specific case. For the previously mentioned case, an average degradation of about 0.4%/°C for Si-based PV technologies was considered which was crucial in order to be able to obtain economic evaluation. The values that were not reported in the examined research studies are marked as "N/A".

The majority of the research studies were related to water based cooling techniques, i.e. more than 65% of them, then air based cooling options, and finally the less investigated cooling options with nanofluids (as the assumed coolant).

Furthermore, the most common obtained studies are ones related to field investigations, which for example is not the case for passive PV cooling options where there is approximately the same number of field and laboratory (or numerically) based investigations [42]. When the application aspects were analysed, it is noticeable that PV/T configurations were the most examined and that CPV configurations were less investigated. Regarding economic aspects, around one quarter of studies analysed them in a particular manner (partial economic evaluation was obtained in most cases), although only one study specified the LCOE for the proposed cooling configuration. LCA aspects were missing and it was not included in any of the herein considered references. Thus, a gap obviously exists regarding the existing literature and both economic and LCA aspects of the proposed active cooling techniques for PVs. The same finding exists for the passive cooling techniques [42].

A summary of the general performance parameters is presented in Table 2 for the herein analysed active cooling options and they are classified according to coolants (water, air and nanofluids). Additionally, as we noticed that PV/T systems are significantly present in the obtained research studies, they were evaluated separately and listed in Table 2 as well. If the data in Table 2 are analysed, it is reasonable to expect an average reduction of PV panel operating temperature in a range of about 10 °C to 20 °C for water based active cooling with an average increase in efficiency ranging about 10% to 20% on average. Furthermore, for PV/T configurations, the average increase in efficiency would range from about 10% to 15% and PV panel operating temperature reduction would range from about 10 °C to 20 °C. In the case of air based cooling, it is reasonable to expect an average increase in efficiency from about 4% to 6% as well as an average reduction of the operating temperature from about 10 °C to 15 °C. Finally, in the case of nanofluids, it was hard to provide an expected efficiency improvement

Table 1
Overview of general data as well as influential parameters for considered active cooling techniques for PVs.

Ref.	PV technology	Coolant/system	Efficiency increase	Increase in the panel power output	Reduction in the panel operating temperature	Economic aspect	Location/solar irradiation level/suns	Other general circumstances
[12]	Si	water	14%	10.5%	22 °C	Yes	Rio De Janerio (Brazil)/N/A	field exp.
[13]	Si (45 W)	air (PV/T)	4%	N/A	5 °C to 10 °C	N/A	Patras (Greece)/200–700 W/m^2	field exp.
[15]	Si	water (PV/T)	2%	N/A	N/A	N/A	Pisa (Italy)/400–1000 W/m^2	num. and field exp.
[17]	Si-mono	water (PV/T)	−10% to −23%	N/A	N/A	N/A	Hefei (China)/300–800 W/m^2	field exp.
[19]	Si-poly (55 W)	air (PV/T)	50% to 55.5%	N/A	30 °C	N/A	Singapore/500–1100 W/m^2	field and num. exp.
[20]	Si-mono	water (CPV)	−22%	16% to 33%	−3°C	N/A	Nanjing (China)/750–1019 W/m^2	field exp.
[21]	Si-poly (55 W)	water (PV/T)	22.2%	25%	10 °C	N/A	Cairo (Egypt)/450–850 W/m^2	field exp.
[22]	Si (180 W)	water (BPVT)	10.7% to 15.1%	N/A	10 °C to 20 °C	Yes	850–1100 W/m^2	laboratory exp.
[23]	Si-mono (230 W)	water	15.5%	10%	7.3 °C to 17 °C	N/A	Dhahran (Saudia Arabia)/750–979 W/m^2	num. and field exp.
[24]	Si-mono (185 W)	water	14.3% to 37.5%	N/A	10 °C to 30 °C	N/A	Cairo (Egypt)/600–800 W/m^2	num. and field exp.
[25]	Si-mono (190 W)	water (PV/T)	14.6% to 18.3%	N/A	33 °C	N/A	Conya (Turkey)/1050 W/m^2	field exp.
[26]	Si	water	5%	N/A	12.5 °C	N/A	Bucharest (Romania)/850–1000 W/m^2	field exp.
[27]	Si-poly (5 W)	nanofluids	7% to 10%	20.57% to 36.67%	18 °C to 24 °C	N/A	1000 W/m^2	laboratory exp.
[28]	Si-mono (40 W)	nanofluids (PV/T)	19.5%	N/A	19 °C	N/A	Mashhad (Iran)/600–1100 W/m^2	field exp.
[29]	Si-poly (5 W)	air	5–10%	3% to 15%	16 °C to 26 °C	N/A	570–1100 W/m^2	num. and laboratory exp.
[30]	Si-mono (0.14 W)	oil (CPV)	−9%	negligible	N/A	N/A	China (Tianjin)/700–925 W/m^2	field exp.
[31]	Si-mono (20 W)	methanol (PV/T)	9% (average)	5.6% to 7.7%	13 °C (average)	N/A	Mashad (Iran)/990–1080 W/m^2	field exp.
[32]	Si-mono (175 W)	water (CPV)	6% to 16%	5% (average)	15 °C to 39 °C	Yes	Portland (USA)/N/A	field exp.
[33]	Si	air (PV/T-PCM)	4% to 8%	N/A	10 °C to 20 °C	N/A	Sydney (Australia)/N/A	num. and field exp.
[34]	Si (250 W)	water	8.3% (average)	15%	12.5 °C to 18.5 °C	Yes	380–798 W/m^2	numerical study
[35]	Si-mono (3.42 W)	water	27.5% to 36.1%	26.1% to 35.5%	12 °C to 26 °C	Yes-LCOE	Dhahran (Saudia Arabia)/400–960 W/m^2	num. and field exp.
[36]	Si-poly (20 W)	water	2.1% to 4%	N/A	0.7 °C to 7.7 °C	N/A	Sharjah (UAE)/849–863 W/m^2	field exp.
[37]	Si-mono (50 W)	water	12% to 14.3%	14% to 16.3%	23 °C to 32 °C	Yes	Split (Croatia)/810–850 W/m^2	field exp.
[38]	Si-mono (50 W)	water	12% to 16%	19% to 25%	19 °C to 26 °C	N/A	Keszthely (Hungary)/N/A	field exp.
[39]	Si (10 W)	air/water	2.6% to 3.0%	8.4% to 13.9%	5 °C to 8 °C	N/A	Beijing (China)/858–1001 W/m^2	field exp.
[40]	Si-poly (40 W)	PCM/water	1.3%	6%	10 °C to 19 °C	Yes	UAE (Al Ain)/970 W/m^2	field exp.
[41]	Si-poly (240–280 W)	air	6.3%	15%	16 °C to 26 °C	N/A	Cartagena (Spain)/1000 W/m^2	field exp.

Table 2
Summary of main performance parameters for herein considered active cooling techniques for PVs.

Cooling technique	Efficiency increase	Increase in panel power output	Reduction in panel operating temperature
water	2.1% to 37.5%	10% to 35.5%	0.7 °C to 32 °C
water (PV/T)	2% to 22.2%	N/A	10 °C to 33 °C
air	2.6% to 10%	3% to 15%	5 °C to 30 °C
nanofluids	7% to 19.5%	20% to 30%	18 °C to 24 °C

range as well as a PV panel operating temperature reduction as only two studies were examined; a representative set of data is missing in order to be able to derive relevant conclusions. However, the specified data in Table 2 shows the respectable potential of nanofluids in actively cooling PVs as they could certainly be competitive when compared to water based active cooling options for PVs.

Therefore, according to the obtained performance analysis, it can be concluded that the water based active cooling strategy is the most viable when efficiency improvement is considered. The air based cooling technique also shows potential; however, it all finally depends from the specific cooling configuration, i.e. its economic viability. Based on the previous summarized performance characteristics, a general economic evaluation was obtained for water and air based active cooling options (i.e. for reasonable chosen cooling concepts that are applicable to the assumed case study of a 30 kW PV system).

4. General economic evaluation: case study of 30 kW PV system that operates in Mediterranean climate conditions

Active cooling techniques are rather different when their economic aspect is analysed. Namely, besides the initial investment, an additional operation (fuel) cost is present in order to enable the cooling technique operation (which strongly affects economic viability). On the other hand, active cooling techniques can boost the desired PV panel's cooling effect and additionally enable the usage of waste heat as it is the case in PV/T configurations. Thus, for active cooling techniques, it is crucial that sufficient energy can be ensured as well as an energy surplus, i.e. reasonable from an economic point of view.

For the purpose of economic evaluation regarding the considered active air or water based cooling technique, the existing 30 kW PV system case study was considered that operated in typical Mediterranean climate conditions, City of Split (Croatia), Fig. 30. The base data related to the PV system are specified in Table 3 and the PV system was monitored via SUNNY PORTAL (web interface). The average annual expected delivered electricity is about 42,000 kW h/year (three year average), with an O & M cost of about 660 €/year. The overall investment into the PV system was 49,700 € and the system would be paid back in about 4 years (the payback period with included governmental subsidies in the annual amount of 13,500 €/year on average). Without governmental subsidies, the expected payback time would range from 8 to 9 years for the specific geographical location. However, the economic analysis in this study was obtained for a case without the consideration of governmental subsidies (general trend for all European countries and as PVs are increasingly competitive [43]).

The general LCOE calculation framework was applied where the LCOE was calculated as follows [44],

$$\mathrm{LCOE} = \frac{\mathrm{AC}}{\mathrm{EO}} = \frac{\mathrm{IC}\times\mathrm{CRF}+\mathrm{OM}}{\mathrm{EO}}, \quad (1)$$

where the capital recovery factor (CRF) was calculated as follows,

$$\mathrm{CRF} = \frac{(1+p)^n \cdot p}{(1+p)^n - 1}. \quad (2)$$

Finally, for the taken interest rate of about 6% and with an amortization period of 30 years, the LCOE was calculated to 0.102 €/kW h for the considered 30 kW PV system and above specified general LCOE calculation framework. The previous result is in accordance with the LCOE reported data by Fraunofer ISE [45], where the LCOE data are specified for current market available PV technologies, and that ranges from 0.08 €/kW h to 0.14 €/kW h.

To obtain an overall economic evaluation, air and water based active cooling techniques were considered as well as reasonable technical solutions (in practical terms) which are based on the existing experimental experience for the herein reported and analysed research studies. Cooling options with nanofluids were not considered due to the general lack of data as well as the lack of research studies to have relevant and referent data for LCOE calculation. In situations where certain data were not available regarding the existing research data that are important for the calculation of LCOE, reasonable data were taken from other literature sources which are related to overall investment or to other details in general.

Finally, the main idea of this section was to check the general economic viability of certain active cooling techniques applied on the realistic case of a 30 kW PV system that operates in Mediterranean climate conditions. As it was hard to estimate the exact realistic operation cost for each considered cooling option (due to numerous influential factors), LCOE was calculated with exempted operation cost (i.e. fuel cost) for each analysed cooling option. Namely, the magnitude of the LCOE is extremely sensitive on the size of the operation cost associated with the nature of the specific considered cooling technique. The most precise way to detect the operational cost is through an experimental approach, where the exact amount of wasted energy (need for cooling technique operation) can be determined by field readings (measurements). In other cases, there is a high chance of gaining misleading results regarding the operation cost estimation for specific cases which we wanted to avoid. Due to the previous reasons, only a few references were found regarding the evaluation of the economic aspect for specific cooling techniques. In those cases, i.e. when the economic aspect was addressed, it was obtained either in the form of an overall investment estimation regarding the cooling technique, or as an overall economic aspect that was based on obtained field measurements. For example, in the reported study [12], the economic aspect was detected with obtained measurements where the proposed techniques turned out not to be economically viable due to inefficient circulation pumps. In paper [22], an overall installation cost was specified in detail (exact cost for each component) and finally the operation cost was calculated based on the experimental experience from the prototype plant. The same approach was applied in research report [32] where the operational cost for the proposed active cooling technique was directly calculated from available measurement data related to the consumed electricity from the circulation pumps. In paper [40], the authors reported the measured consumed energy needed for the operation of the



Fig. 30. Photo of considered 30 kW PV system (University of Split).

Table 3
Base data related to considered 30 kW PV system.

General data for 30 kW PV system	
Photovoltaic technology	Si-polycrystalline
Overall nominal power output	30 kW
Nominal panel power output	250 W
Overall PV panel surface	204.6 m^2
Nominal efficiency	15.37%
Number of PV panels	126
Overall investment	49,700 €
Average produced electricity	42,070 kW h/year
Annual O & M cost	660 €/year

considered cooling technique. Thus, from the previous examples found in the available existing literature data, it can be noticed that the operation cost for specific active cooling techniques was strictly reported based on direct experimental experience (measurements).

The herein applied approach is rather reasonable due to the previously addressed issues and it provides useful data related to the initial economic viability of the examined cooling options, where the later operation (fuel) cost can be considered and added (based on the measurements).

Finally, it needs to be emphasized that our intention was to obtain a general economic evaluation and to get the initial economic starting position for each active cooling technique, which is also useful information (but it is this study's limitation). However, to be able to get a more precise economic aspect, the optimization of each considered cooling technique is crucial as more realistic economic indicators would be ensured in that case (and the LCOE would then be more reduced).

4.1. Air based active cooling technique

When analysing the rate of efficiency improvement gained in the obtained and herein reported research studies, where active air based cooling was assumed, it is reasonable to expect an average increase at least of 7% in delivered electricity (which is a rather conservative approach but reasonable). In general, the average efficiency increase would range somewhat between at least 5% to maximal 10%, which was based according to the provided existing research findings review (a specified performance range was used for the calculation of the LCOE data specified in Table 4). Furthermore, for the considered 30 kW PV system, it is assumed that the PV panel would be cooled by a blower fan and induced airflow stream on the backside surface of the PV panel with an included duct system to direct airflow more efficiently (similar approach as presented in [41]) and a metal stand to mount the blower fan. For the existing 30 kW PV system, each 250 W PV panel would annually deliver about 334 kW h/year of electricity, and in the case of the air induced cooling, it would deliver about 357 kW h/year (i.e. the whole PV plant would deliver about 45,015 kW h/year of electricity). Further, by applying the active cooling technique, it is reasonable to expect an increase in PV system lifetime to about 40 years [46]. Thus, if we take all the previously mentioned data into account, as well as the performance data in Table 2 to ensure minimal LCOE, the maximal investment regarding the considered cooling system could be around 6300 €, i.e. the allowed additional investment into the considered cooling option is around 30 €/m^2 (and which leads to an LCOE of about 0.097 €/kW h). The previously specified maximal investment of 6300 € was set in order to get a reasonable calculated LCOE of 0.097 €/kW h, where the overall investment should include the cost of a 2.0 mm thick aluminium sheet (around 2.0 €/kg [47]), the cost for a metal stand, fan blower and labor cost (about 15–20 €/h, which strongly depends from the specific country).

Assuming that the cost of a fan blower would certainly range from 50 € to 100 € per PV panel [48], it leads to an additional investment scale of about 31 €/m^2 to 62 €/m^2 for PV panels. Taking into account the average expected increase of about 7% in performance and the previously specified expected overall cost, the LCOE would range from 0.101 €/kW h to 0.11€/kW h, for an estimated O & M cost in the amount of 800 €/year. However, it needs to be emphasized that in the previous calculation, the cost for consumed electricity to drive fans was not included as our idea was to initially check economic viability in the first approach. Namely, if we assume that each blower fan would have about 80 W of nominal power output with an operational time of about 8 h/day minimum, the roughly overall consumed electricity for a single blower fan would be about 0.65 kW h/day, i.e. annually around 230 kW h/year. It was already specified above that the overall performance benefit in delivered electricity is estimated to be around 24 kW h/year per PV panel, even from rough calculations, it is clear that economic viability does not exist for the examined air based active cooling option. Further, it is certainly reasonable to expect the O & M cost to be higher than the above specified due to a large number of fans and their general sensitivity regarding the working circumstances which finally leads to a higher LCOE value (it is hard to specify the exact value but the situation is certainly clear).

Thus, based on the previous calculations, it is clear that the air based cooling option (blower fan with a specific air duct system) is not an economically viable solution for the 30 kW PV system case. Namely, the overall investment for the considered cooling technique is relatively high and the performance benefit is not sufficient.

4.2. Water based active cooling techniques

4.2.1. PV/T configuration

According to the herein obtained review of the existing research findings, the water based cooling techniques are certainly more efficient ones when compared to air based cooling techniques and it is reasonable to expect the range of overall efficiency improvement to be from 15% to 20% on average. Taking into account the previously specified data for the efficiency improvement, as well as the data in Table 2, the overall additional investment for modifying the existing 30 kW PV plant into a PV/T configuration would range from 11,300 € to 14,300 €, with minimal economic benefit, i.e. set LCOE value of 0.097 €/kW h (the previously specified estimated overall allowed investment was calculated with the general limit to achieve an economically reasonable LCOE and which was calculated to the already specified value 0.097 €/kW h). Therefore, the further point is to check if the proposed modification from PV to PV/T configuration is possible to achieve at all, i.e. to be able to fit the overall cost within the above-specified expected allowed overall investment (one that will ensure reasonable LCOE). Scaled on the overall PV panel surface, it leads to the overall allowed investment of about 55 €/m^2 to 70 €/m^2 (i.e. for a 250 W PV panel from 90 € to 114 €) to modify the existing PV system to PV/T. Further, the average investment in market available PV/T configurations is about two to three times higher when compared to single PV panels. Namely, the average retail price of the considered 250 W PV panel is about 0.85 €/W, i.e. about 212 € for the whole PV panel. Thus, the difference in initial investment between a single PV panel and PV/T panel ranges from about 0.85 €/W to 1.7 €/W (i.e. scaled on the whole PV panel, it is from 212 € to 425 €). As previously estimated, the additional allowed overall investment to obtain a PV/T configuration

Table 4
General LCOE data for considered 30 kW PV system.

General LCOE (€/kW h) data for the 30 kW PV system	
Referent PV system	0.102
Air based cooling	0.101–0.11
Water (PV/T configuration)	0.124–0.159
Water (spray cooling)	0.103–0.108
Water (front water layer)	0.096–0.10

ranges from 55 €/m^2 to 70 €/m^2 (to obtain minimal LCOE), it is certainly clear from an economic point of view that there is no sense to invest into a PV/T configuration for the given circumstances and assumed modification. Currently, the initial investment regarding PV/T configurations is relatively high and the performance improvement is not sufficient. However, the additional benefit from a PV/T configuration would be in the availability of hot water utilization and should be taken into account. In fact, the main point of a PV/T system is to ensure both electricity and hot water (which could additionally help improve economic feasibility, i.e. to reduce LCOE. Namely, with the assumed PV/T configuration (i.e. modification of existing PV system to PV/T configuration), it was calculated that the system would able to ensure a thermal output of about 130.000 kW h/year (based on the average thermal efficiency of about 40% for the PV/T system). Currently, in the case of the University building, hot water is prepared with electric boilers, thus the PV/T configuration would lead to a favourable economic effect. The calculated amount of produced thermal output from the PV/T system would ensure about 7700 €/year in annual savings regarding spent electricity. Taking into account the average expected investment of about 318 €/m2 for a 30 kW PV system regarding the modification of PV to PV/T configuration (average value taken based on the above specified range), it would lead to an overall investment of about 65.000 € (with the included PV system investment). Thus, with the previously estimated savings regarding hot water preparation (7700 €/year), the simple payback period would be about 8.5 years (simple payback period is expected to be somewhat higher due to a labor cost that was not included in the previous rough calculation). Further, when the LCOE was calculated and scaled on the overall produced energy output from the PV/T system (electricity hot water), with the expected lifetime of about 20 years for the PV/T system, the LCOE was calculated to about 0.064 €/kW h (O & M cost was set to 1000 €/year, but with no included costs for the water circulation system). Thus, the general economic benefit of the PV/T configuration is clear when scaled on the overall produced energy output. However, this study was limited to an electric aspect analysis and not thermal as by including the thermal aspect, it would lead beyond the main scope of the herein investigated topic. Additionally, one of the issues associated with the PV/T configuration is the general lifetime that is usually specified by the producers and limited to around 15 years. The PV system's lifetime would certainly be prolonged to 25 or 30 years with the application of the proposed cooling technique, and that is also one of the reasons why it is hard to provide a fair economic comparison of the herein considered options. Finally, the economic viability of the PV/T configuration was certainly showed for the specific case but due to the general limitation of this study, as well as other addressed general reasons above, PV/T systems were not analysed in the continuation.

4.2.2. Liquid based cooling technique over front PV panel surface

By applying liquid based cooling over the front PV panel surface, it was found that the average operating temperature of the PV panel can be reduced by 20 °C to 30 °C [12,24,37]. Furthermore, a second benefit could be ensured with the regular cleaning of the front PV surface and would cause an additional increase in the annual delivered quantity of electricity from the PV system. The expected corresponding increase in efficiency by applying the proposed active cooling technique would range from about 10% to 15% on average. A suitable and reasonable liquid based cooling technique could basically be enabled either through the application of a water layer over the front PV panel surface (via a drilled water tube system [12]) or through a water nozzle system with applied water spray over the PV panel surfaces [37]. In both cases, a water reservoir is needed for the circulating water and even water cooling in some circumstances (in order to ensure a stable temperature of cooled water). The cooling of water could be an issue in regions with high solar irradiation levels due to intense evaporation and a relatively significant magnitude of waste heat removed from the PV panel in general. However, the optimal water temperature as well as duration of the applied cooling should be determined for the specific case taking complex circumstances into account as well as economic viability for the assumed cooling. Namely, the gained performance benefit can be annulled due to the cooling of PV panels and the required energy consumption for the circulation system as well as the additional water cooling (if it exists, we did not assume the cooling of water in further analysis). A crucial feature related to both mentioned techniques, is the quantity of consumed energy for the water circulation system. Further, it is clear that we are dealing with a sensitive energy balance system that requires a quality enabled control, i.e. a regulation system in order to ensure an economically reasonable cooling technique.

The advantage of the water spray based cooling techniques is that it could be applied simultaneously over the front and backside of PV panel surfaces which will finally result in higher performance and a more efficient reduction rate of the PV panel operating temperature, when compared it with front PV cooling only with the application of a water layer. Further, the evaporation issue could be important as a certain quantity of water will certainly evaporate in both cases (the magnitude of the evaporation factor strongly depends from the general climate conditions, i.e. solar irradiation levels, partial pressures, latent heat of evaporation, etc.). Additionally, it is worth emphasizing that water is a valuable and limited resource, so in any possible case, rainwater could be a more reasonable option than pipeline water, but finally it all depends from the other general circumstances and economic outcome. In the reported study [12], the average experimentally detected hourly evaporation rate was of around 1.0 l/h per m^2. Therefore, water evaporation is certainly an influential factor that can affect overall cooling system benefit and it generally depends from a variety of influential factors.

The O & M cost for the considered systems could also be an issue due to a relatively complex pipeline system and general sensitive equipment when exposed long-term to sun irradiation (and which will certainly cause occasional additional investments in reparations and general cooling system maintenance).

Regarding the overall investment into the cooling system (water spray cooling technique), i.e. according to the experience in [37], it is reasonable to expect an overall investment cost of about 50 €/m^2 for the assumed constructional parts related to the considered water spray based cooling technique (water nozzles, connection elements, required pipeline armature, circulation pump, water reservoir and mounting system, where the average prices for the specified elements can be found in [49]). Thus, if we take the expected average increase in performance to range from 10% to 15% and the expected overall cost in the amount of about 60,000 €, the calculated LCOE would range from about 0.103 €/kW h to 0.108 €/kW h. The previous LCOE was calculated for an estimated O & M cost of around 1000 €/year but without the included cost of consumed electricity needed to enable water circulation in the system (this kind of approach was used for initial economic evaluation as already emphasized). Therefore, from an economic point of view, the calculated LCOE is clearly high, as it does not include the consumed electricity for the water circulation system. Finally, the examined water spray cooling technique could only be attractive in the case of a significant overall investment reduction (roughly max 20 €/m^2) together with a reasonable O & M cost (for the considered case not more than 700 €/year) that would be ensured.

The second water based cooling option, i.e. application of a water layer over the front PV panel surface is significantly lower when compared to the overall investment and is estimated not to exceed 25 €/m^2 (the cost includes polyurethane water pipelines, duct elements, circulation pumps, where the general average cost of each element can be explored in [50] and which was also used in this study), which leads to an overall investment of about 54,900 €. Taking all the same above specified general circumstances into account, the LCOE was calculated to range from 0.096 €/kW h to 0.10 €/kW h and is obviously not an economically viable option (if the additional cost for the water

circulation system was added, the LCOE would certainly be higher). Thus, all the above considered water based cooling techniques in this section are critical from an economic point of view but on the other hand have definite potential that could be ensured by optimizing the cooling technique in order to reduce the operating cost and finally lead to a more attractive LCOE.

4.3. Summary

According to the above obtained analysis related to the general economic aspect of the assumed 30 kW PC system, the LCOE was summarized in Table 4.

However, it is once again important to emphasize that the above specified LCOE would additionally be increased due to the present cost of consumed energy for the cooling system's operation (the LCOE for the PV/T configuration was calculated and only scaled on the electricity aspect, i.e. thermal output was not included). The main point of the herein obtained economic evaluation was to show the realistic potential of the considered active cooling techniques and the assumed case study of a 30 kW PV system. From the above gained data, it is clearly reasonable not to invest in all the examined cooling techniques as they are not economically viable options in this specific case (front water layer cooling options have an economic potential but are strongly sensitive to O & M cost rates). In fact, for the herein studied case, the active cooling of PVs could only be economically viable in the case of removed heat utilization from the PV panel (i.e. in the form of hot water which would finally ensure additional economic benefit and an economically more reasonable cooling option). As previously emphasized, optimization is crucial for active cooling techniques and it should be followed with the "xsmart" regulation operation of the system (in order to ensure reasonable LCOE).

Thus, PV/T configurations are the only reasonable ones currently (due to additional economic benefit related to hot water) when compared to pure active cooling techniques for PVs without the considered utilization of waste heat from the PV panel.

5. Environmental aspect of active cooling techniques: LCA analysis

Europe is facing challenges in the energy and financial sector. The unstable economic circumstances are an intense source of disturbance on the continued growth of European countries. The ongoing increase of energy demands, observed in past years, led to the gradual depletion of fossil fuels. Furthermore, the significant increases of gas pollutants (CO_2, NOx) have had major impacts on the climate and environment (eg. global warming) [51].

This paper also targets evaluating the environmental impact of the suggested cooling techniques by using the LCA approach. A life cycle assessment was performed following the international ISO 14040 series standards [52–55] supported by Gabi Software [56]. The aim was to trace the system's environmental impact related to its life cycle. When thinking of Life cycle in environmental impact assessment terms, it has now been prevalent for over 40 years in the industrial and service sector, with some early Life Cycle Analysis (LCA) type practitioners bringing a multi-attribute quantitative approach to decisions related to beverage packaging as far back as the 1960s. This approach evolved into LCA, with well-established international standards for guidance. LCA is therefore a science-based, quantitative and integrative methodology that measures the material and energy flows to and from nature over the lifetime of a product or service (and assesses the potential impact of those flows on resources, ecosystems and human health). These impacts occur in various segments of the manufacturing value chain and throughout the life of a product (carbon footprint, acidification, eutrophication, photochemical oxidant creation and ozone depletion).

The main purposes of LCA are to identify hot spots in the value chain, identify and quantify alternatives, and disclose environmental information. The following phases have been evaluated: production and transportation of energy and raw materials, production processes, installation, maintenance, disposal and transports occurring during each step. The LCA approach used is best termed as "streamlined" LCA (SLCA), as it does not take into account the recycling or the disposal into landfill. The main purpose of planning is to define the scope and goal as well as the boundaries of the system studied. During the planning, the objective should be clear and the data collection sources defined. The inventory analysis is the stage where the inputs and outputs are quantified. Energy, raw materials, water consumption, air emissions, solid waste, are quantified either through measurements or database searches, surveys and software calculations.

During impact assessment, the environmental impact is calculated as a result of inventory analysis. The input and output data are translated into environmental impacts like climate change, acidification, eutrophication, and photochemical oxidation. Finally, the stage of improvement analysis involves discussion and improvement suggestions. For implementing LCA at the inventory phase, a software tool was used for the results' reliability control: the GABI LCA software, which is a life-cycle analysis model with an embodied EcoInvent LCA. At the environmental impact assessment phase (normalization and weighting) a set of specialized indicators were used, derived from CML 2.

CML 2001 is an impact assessment method which restricts quantitative modelling to early stages in the cause-effect chain to limit uncertainties. The results are grouped in midpoint categories according to common mechanisms (e.g. climate change) or commonly accepted groupings (e.g. ecotoxicity). CML 2001 is developed by the Institute of Environmental Sciences, Leiden University, The Netherlands, and is published in a handbook with several different authors [57]. The main principles behind the methodology are not being further developed. A Microsoft Excel spreadsheet with characterisation factors for more than 1700 different flows can be downloaded from the CML website. The characterisation factors are updated when new knowledge on a substance level is available. Several additional characterisation factors are calculated by PE and LBP-GaBi following the principles described in the CML 2001 methodology documents. The normalization factors for CML 2001 are available for the Netherlands, Western Europe, EU and the World. The normalization factors are calculated via total substance emissions and characterisation factors per substance, and hence follow the substance level updates as described above. The normalization data for other countries (e.g. Germany, USA and Japan) and geographical units (e.g. North America or OECD) are available in GaBi5.

The analysis was used as a tool to define the energy consuming processes taking into consideration the most significant energy consumptions and improve the present situation of the system studied. Energy and environmental goals were set to improve the environmental and consequently the energy efficiency processes.

The three active cooling techniques were compared and analysed from an environmental point of view (i.e. ones that were previously evaluated). The specific input data for the considered cooling techniques are presented in Table 5. For each cooling technique, the quantity of used constructional materials was estimated, i.e. the main components of the system and the considered 30 kW PV system as the case study base (the specific available data from the Refs. [12,19,24,29,37,41] were used and applied on examined PV system for estimation of the constructional materials used).

According to the obtained LCA analysis, it is obvious from Fig. 31 that Option I, i.e. the air based cooling technique has the most significant environmental impact, especially when considering global warming and acidification impacts. This is caused because of the aluminium sheet metal and the axial blower fan which is mainly made of steel.

Nevertheless, the environmental impact caused by option I is not so significant compared to the other options, especially for environmental impacts such as eutrophication, ozone layer depletion, abiotic

Table 5
General input data used for LCA related to considered specific cooling techniques.

Option I – Air based cooling technique over backside PV panel surface:
Al sheet metal (1.0 mm): 315 kg Axial blower fan (300 mm diameter), nominal power 80 W Iron stand (for axial blower fan): 400 kg
Option II – Liquid based cooling technique over front PV panel surface:
PVC quantity (pipes and armature in general): 150 kg Water storage tank: 2 × 1 m^3 (two tanks) Circulation pump: 1.5 kW, H = 50 m
Option III – water spray cooling technique over front PV panel surface
PVC quantity (pipes and armature in general): 210 kg Aluminium wire (1.5 mm): 80 kg Water storage tank: 2 × 1 m^3 (two tanks) Circulation pump: 1.5 kW, H = 65 m

depletion, human toxicity and photochemical ozone depletion, Table 6.

In conclusion, it could be said that the environmental parameters are not the criteria that will actually influence the decision making process dramatically but should more importantly lean on the economic aspect and general technical viability of the proposed cooling techniques in practical terms.

6. Conclusions

Active cooling techniques for photovoltaics were analysed in this paper where the integral approach was applied to assess performance, economic and environmental aspects. An economic and environmental analysis was obtained for the case of a 30 kW PV system to be able to get a realistic outcome and check suitability of the cooling options from a practical point of view. According to a performed and detail comprehensive review of active cooling techniques, for the majority of the herein analysed studies, water was used as the coolant and in PV/T configurations mostly. The second most used coolant was air and finally nanofluids which are less investigated.

Table 6
Environmental impact of different cooling techniques.

Environmental impact		Option I	Option II	Option III
Global Warming	tn CO_2 – eq	33.74028	31.64988	32.28388
Acidification	kg SO_2 – eq	139.2556	124.539	127.943
Eutrophication	kg PO_4 – eq	19.59108	18.95546	19.13426
Ozone Layer Depletion	kg CFC – eq	0.008813	0.008691	0.008732
Abiotic Depletion	kg Sb – eq	23.9409	23.94129	23.94196
Human Toxicity	tn Db – eq	18.2467	17.61266	17.77336
Photochemical Ozone Depletion	kg C_2H_4 – eq	7.69126	7.0058	7.2748

Regarding the expected performance benefit of the herein considered active cooling techniques, the average improvement in performance would range from about 10% to 15% for active water based cooling techniques and about 5% to 10% for air based cooling techniques. However, it needs to be emphasized that the previously specified data are not net values (i.e. realistic performance benefit would be lower due to energy consumption needed for the operation of the cooling system). Nanofluids are found to be promising coolants regarding their potential performance benefit and reduction in operating temperatures. Further, there are just a few studies that considered nanofluids and in general a representative number of studies are missing to derive respectable conclusions (and especially their economic evaluation). Thus, cooling with nanofluids should be investigated more intensely for PV applications.

For the considered 30 kW PV system, it was found that only water based cooling is reasonable from a technical point of view in two variants (water layer over the front PV surface or application of water spray). However, when analysing the economic aspect, it was found that the herein examined active cooling techniques are questionable,

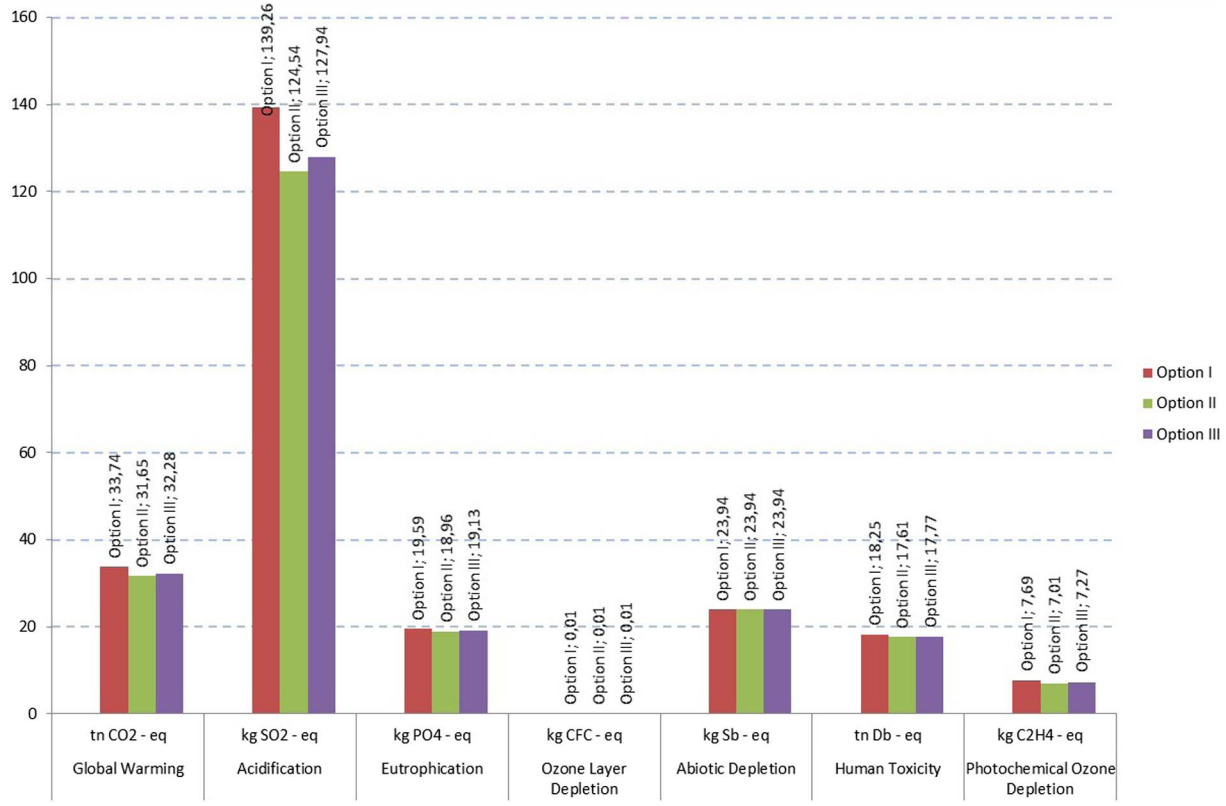


Fig. 31. Environmental performance of different cooling techniques based on LCA methodology.

and in fact they are currently not economically viable. Namely, it needs to be said that initial evaluation was obtained by not considering the consumed energy for the operation of the specific cooling system which would significantly reduce the gained benefit due to enabled cooling and increase LCOE. However, that approach was reasonable and sufficient as it provided general insight into the base economic viability of the herein chosen cooling techniques. On the other side, PV/T configurations can bring additional economic benefit due to the utilization of waste heat (additional decrease of LCOE), but improvements in delivered electricity were examined so the thermal aspect was not included.

Regarding environmental aspects, air and water based cooling techniques were compared through conducted LCA analysis. The gained results showed that the most harmful cooling technique is the air based technique as it has the highest environmental impact when compared to other examined cooling techniques (the most emphasized environmental impacts are global warming and acidification effect). Other impacts are more or less similar for all the herein considered cooling techniques. Thus, as already addressed, environmental impacts should not be crucial for the selection of the specific cooling technique.

According to the herein obtained analysis of the active cooling techniques for photovoltaics and considered case study (specific Mediterranean climate conditions), it can be concluded that water based cooling should be considered first as a viable cooling option. However, in order to ensure more reasonable LCOE, further optimization of the existing cooling methods, or even development of the novel cooling techniques, should be obtained to ensure additional increases in performance and enhance the benefit of cooling. The previous should be followed with the "smart" regulation system for specific assumed active cooling techniques in order to be able additionally to reduce LCOE. Further, LCOE could be also reduced with the efficient utilization of waste heat and which should be mandatory for active cooling techniques. The self-cleaning effect on the front PV surface can also bring extra benefit in performance, thus all previously mentioned issues could have a favourable effect on the LCOE. Finally, in the near future, active cooling techniques will certainly have their place on the market due to present and upcoming climate issues as there will be a struggle to remain reasonably efficient siliceous-based PV technologies in the circumstances of the increased surrounding air temperatures (and not to mention that Si-based PV technologies have the highest market share today).

References

[1] Parida B, Iniyan S, Goic R. A review of solar photovoltaic technologies. Renew Sustain Energy Rev 2011;15:1625–36.

[2] Web source; 2017. < http://www.renewableenergyworld.com/articles/2014/01/bifacial-pv-modules-can-they-move-beyond-bipv-applications.html > [last visited, April 18, 2017].

[3] Web source; 2017. < https://panasonic.net/ecosolutions/solar/ > [last visited, April 18, 2017].

[4] Web source; 2017. < https://www.ise.fraunhofer.de/content/dam/ise/de/documents/publications/studies/Photovoltaics-Report.pdf > [last visited, April 19, 2017].

[5] Nižetić S, Duić N, Papadopulos AM, Tina GM, Grubišić-Čabo F. Energy efficiency evaluation of a hybrid energy system for building applications in a Mediterranean climate and its feasibility aspect. Energy 2015;90:1171–9.

[6] Tiwari GN, Mishra RK, Solanki SC. Photovoltaic modules and their applications: a review on thermal modelling. Appl Energy 2011;88:2287–304.

[7] Nižetić S, Grubišić- Čabo F, Marinic-Kragić I, Lastname FN. Experimental and numerical investigation of a backside convective cooling mechanism on photovoltaic panels. Energy 2016;111:211–25.

[8] Daghigha R, Ruslana MH, Sopiana K. Advances in liquid based photovoltaic/thermal (PV/T) collectors. Renew Sustain Energy Rev 2011;15:4156–70.

[9] Al-Shamani Ali Najah, Yazdi Mohammad H, Alghoul MA, Abed Azher M, Ruslan MH, Mat Sohif, et al. Nanofluids for improved efficiency in cooling solar collectors – a review. Renew Sustain Energy Rev 2014;38:348–67.

[10] Arcuri Natale, Reda Francesco, De Simone Marilena. Energy and thermo-fluid-dynamics evaluations of photovoltaic panels cooled by water and air. Sol Energy 2014;105:147–56.

[11] Jakhar Sanjeev, Soni MS, Gakkhar Nikhil. Historical and recent development of concentrating photovoltaic cooling technologies. Renew Sustain Energy Rev 2016;60:41–59.

[12] Krauter Stefan. Increased electrical yield via water flow over the front of photovoltaic panels. Sol Energy Mater Sol Cells 2004;82:131–7.

[13] Tonui JK, Tripanagnostopoulos Y. Improved PV/T solar collectors with heat extraction by forced or natural air circulation. Renew Energy 2007;32:623–37.

[14] Tonui JK, Tripanagnostopoulos Y. Performance improvement of PV/T solar collectors with natural air flow operation. Sol Energy 2008;82:1–12.

[15] Rosa-Clot M, Rosa-Clot P, Tina GM. TESPI: thermal electric solar panel integration. Sol Energy 2011;85:2433–42.

[16] Rosa-Clot M, Rosa-Clot P, Tina GM, Scandura PF. Submerged photovoltaic solar panel: SP2. Renew Energy 2010;35:1862–5.

[17] He Wei, Zhang Yang, Ji Jie. Comparative experiment study on photovoltaic and thermal solar system under natural circulation of water. Appl Therm Eng 2011;31:3369–76.

[18] Pellicone Devin, Ortega Alfonso, del Valle Marcelo, Schon Steven. Simulation of two-phase flow and heat transfer in mini and micro channels for concentrating photovoltaics cooling. In: Proceedings of the ASME 2011 5th international conference on energy sustainability ES2011 August 7–10, Washington, DC, USA; 2011.

[19] Teo HG, Lee PS, Hawlader MNA. An active cooling system for photovoltaic modules. Appl Energy 2012;90:309–15.

[20] Du Bin, Hu Eric, Kolhe Mohan. Performance analysis of water cooled concentrated photovoltaic (CPV) system. Renew Sustain Energy Rev 2012;16:6732–6.

[21] El-Seesy EI, Khalil T, Ahmed HM. Experimental investigations and developing of photovoltaic/thermal system. World Appl Sci J 2012;19(9):1342–7.

[22] Yin HM, Yang DJ, Kelly G, Garant J. Design and performance of a novel building integrated PV/thermal system for energy efficiency of buildings. Sol Energy 2013;87:184–95.

[23] Bahaidarah H, Subhan Abdul, Rehman S. Performance evaluation of a PV (photovoltaic) module by back surface water cooling for hot climatic conditions. Energy 2013;59:445–53.

[24] Moharram KA, Abd-Elhady MS, Kandil HA, El-Sherif H. Enhancing the performance of photovoltaic panels by water cooling. Ain Shams Eng J 2013;4:869–77.

[25] Ozgoren1 M, Aksoy MH, Bakir C, Dogan S. Experimental performance investigation of photovoltaic/thermal (PV-T) system. In: EPJ web of conferences 45,01106; 2013.

[26] Dorobanțu L, Popescu MO, Popescu CL, Crăciunescu A. Experimental assessment of PV panels front water cooling strategy. In: International conference on renewable energies and power quality (ICREPQ'13) Bilbao (Spain), 20th to 22th March; 2013.

[27] Karami Nooshin, Rahimi Masoud. Heat transfer enhancement in a PV cell using Boehmite nanofluid. Energy Convers Manage 2014;86:275–85.

[28] Sardarabadi Mohammad, Passandideh-Fard Mohammad, Heris Saeed Zeinali. Experimental investigation of the effects of silica/water nanofluid on PV/T (photovoltaic thermal units). Energy 2014;66:264–72.

[29] Rahimi Masoud, Valeh-e-Sheyda Peyvand, Parsamoghadam Mohammad Amin, Masahi Mohammad Moein, Alsairafi Ammar Abdulaziz. Design of a self-adjusted jet impingement system for cooling of photovoltaic cells. Energy Convers Manage 2014;83:48–57.

[30] Sun Yong, Wang Yiping, Zhu Li, Yin Baoquan, Xiang Haijun, Huang Qunwu. Direct liquid-immersion cooling of concentrator silicon solar cells in a linear concentrating photovoltaic receiver. Energy 2014;65:264–71.

[31] Moradgholi Meysam, Nowee Seyed Mostafa, Abrishamchi Iman. Application of heat pipe in an experimental investigation on a novel photovoltaic/thermal (PV/T) system. Sol Energy 2014;107:82–8.

[32] Smith Matthew K, Selbak Hanny, Wamser Carl C, Day Nicholas U, Krieske Mathew, Sailor David J, et al. Water cooling method to improve the performance of field-mounted, insulated, and concentrating photovoltaic modules. J Sol Energy Eng 2014;136. 034503-1.

[33] Lin Wenye, Ma Zhenjun, Imroz Sohel M, Cooper Paul. Development and evaluation of a ceiling ventilation system enhanced by solar photovoltaic thermal collectors and phase change materials. Energy Convers Manage 2014;88:218–30.

[34] Wu Shenyi, Xiong Chenguang. Passive cooling technology for photovoltaic panels for domestic houses. Int J Low-Carbon Technol 2014;1–9.

[35] Baloch Ahmer AB, Bahaidarah Haitham MS, Gandhidasan Palanichamy, Al-Sulaiman Fahad A. Experimental and numerical performance analysis of a converging channel heat exchanger for PV cooling. Energy Convers Manage 2015;103:14–27.

[36] Hachicha Ahmed Amine, Ghenai Chaouki, Hamid Abdul Kadir. Enhancing the performance of a photovoltaic module using different cooling methods. Int J Environ Chem Ecol Geol Geophys Eng 2015;9(9):1106–9.

[37] Nižetić S, Čoko D, Yadav A, Grubišic-Čabo F. Water spray cooling technique applied on a photovoltaic panel: the performance response. Energy Convers Manage 2016;108:287–96.

[38] Bai Attila, Popp József, Balogh Péter, Gabnai Zoltán, Pályi Béla, Farkas István, et al. Technical and economic effects of cooling of monocrystalline photovoltaic modules under Hungarian conditions. Renew Sustain Energy Rev 2016;60:1086–99.

[39] Tang Xiao, Quan Zhenhua, Zhao Yaohua. Experimental investigation of solar panel cooling by a novel micro heat pipe array. Energy Power Eng 2010;2:171–4.

[40] Hasan Ahmad, Alnoman Hamza, Shah Ali Hasan. Energy efficiency enhancement of photovoltaics by phase change materials through thermal energy recovery. Energies 2016;9:782. http://dx.doi.org/10.3390/en9100782.

[41] Mazón-Hernández R, García-Cascales JR, Vera-García F, Káiser AS, Zamora B. Improving the electrical parameters of a photovoltaic panel by means of an induced or forced air stream. Int J Photoenergy 2013, [Article ID 830968].

[42] Nižetić S, Papadopoulos AM, Giama E. Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels. Part I: passive cooling techniques. Energy Convers Manage 2017;149:334–54.

[43] Anastaselos DA, Oxizidis S, Manoudis A, Papadopoulos AM. Environmental performance of energy systems of residential buildings: towards sustainable

communities. Sustain Cities Soc 2016;20:96–108.
[44] Ramadhan M, Naseeb A. The cost benefit analysis of implementing photovoltaic solar system in the state of Kuwait. Renew Energy 2011;36:1272–6.
[45] Web source; 2017. < https://www.ise.fraunhofer.de/content/dam/ise/en/documents/publications/studies/Fraunhofer-ISE_LCOE_Renewable_Energy_technologies.pdf > [last visited April 20, 2017].
[46] Royo Patricia, Ferreira Víctor J, Lopez-Sabiron Ana M, Ferreira German. Hybrid diagnosis to characterise the energy and environmental enhancement of photovoltaic modules using smart materials. Energy 2016;101:174–89.
[47] Web source; 2017. < https://www.alibaba.com/trade/search?fsb=y&IndexArea=product_en&CatId=&SearchText=aluminium+sheet+per+kg+thickness+2mm&isPremium=y > [last visited, August 24, 2017].
[48] Web source; 2017. < https://www.alibaba.com/trade/search?IndexArea=product_en&CatId=&fsb=y&isPremium=y&SearchText=axial+fan+blower+80+W > [last visited, June 20, 2017].
[49] Web source; 2017. < https://www.alibaba.com/trade/search?fsb=y&IndexArea=product_en&CatId=&SearchText=water+spray+nozzle+high+pressure&isPremium=y > [last visited, June 20, 2017].
[50] Web source; 2017. < https://www.alibaba.com > [last visited, June 22, 2017].
[51] Giama E. Energy performance of buildings – energy efficiency and built environment in temperate climates. In: Santamouris M, editor. Life cycle vs carbon footprint analysis for construction materials, New York: Springer Science & Business Media; 2016. p. 95–106. ISBN 978-3-319-20830-5, ISBN 978-3319-20831-2 (eBook). http://dx.doi.org/10.1007/978-3-319-20831-2 [chapter 6].
[52] ISO 14040. Environmental management: life cycle assessment, principles and framework; 1997.
[53] ISO 14041. Environmental management: life cycle assessment, principles and framework; 1998.
[54] ISO 14042. Environmental management: life cycle assessment, life cycle impact assessment; 2000.
[55] ISO 14043. Environmental management: life cycle assessment, life cycle interpretation; 2000.
[56] Web source; 2016. < http://www.gabi-software.com/international/index/ > [last visited, June 15].
[57] Guinée JB, Gorrée M, Heijungs R, Huppes G, Kleijn R, Koning A, et al. Handbook on life cycle assessment. Operational guide to the ISO standards. I: LCA in perspective. IIa: Guide. IIb: Operational annex. III: Scientific background. Dordrecht: Kluwer Academic Publishers; 2002. 692 pp. ISBN 1-4020-0228-9.